# Causal Clifford Theory (CL10): the Standard Model and gravity from a single Clifford root on a Planck-lattice quantum cellular automaton

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We present **Causal Clifford Theory (CL10)**: a unitary quantum cellular automaton (QCA) on a cubic lattice of Planck spacing whose on-site internal space is the 32-dimensional complex module of the Clifford algebra $\mathrm{Cl}(10,0)$. Ten mutually anticommuting Hermitian generators $h_a^2 = I$ supply the kinematics. From this single algebraic object follow a (1+3) Minkowski subalgebra; a 16-component Weyl module whose charge assignment is exactly one Standard Model (SM) generation, with all eight gauge and mixed anomalies cancelling; a 45-dimensional Spin(10) adjoint containing the twelve SM gauge bosons; the 5/3 hypercharge normalisation; an exactly unitary, strictly causal update with $\det U = 1$ and an exact matter–antimatter spectral mirror; and a lattice theory of a massless spin-2 field whose discrete Einstein–Hilbert action reproduces general relativity to machine precision.

We also state **what does not follow**. Every number is classified by provenance — *algebraic*, *conventional*, *fitted*, or *open*. The family number, the Yukawa structure, quark mixing, the cosmological constant, the dark-matter identity and abundance, the exact neutrino mass, the baryon asymmetry and one-loop coupling unification remain open. A single loop factor $(1/16\pi)^{10}$ reproduces the electroweak/Planck hierarchy to a factor 2.08 and gives a 118 GeV dark-matter scale, but the exponent 10 is selected by the data it is meant to explain.

Five tensions with observation are reported rather than hidden. Under Standard-Model desert running the GUT coupling $\alpha_{10} = 1/25$ overshoots the measured SU(2) coupling by a factor 5.8 and drives SU(3) to a Landau pole at $2.2 \times 10^9$ GeV; this failure is repaired by the vectorlike remnant matter the theory's own breaking sector supplies — three $\mathbf{10} \oplus \overline{\mathbf{10}}$ pairs plus one weak-doublet pair make the three measured couplings meet at $\alpha_X = 1/25$ *exactly*, at $M_X = 1.8 \times 10^{16}$ GeV, at the cost of abandoning $M_{\mathrm{GUT}} = M_P$ and of two fitted thresholds; the repair predicts $\tau_p \simeq 1.8 \times 10^{36}$ yr. The remnant is no longer an assumption: an internal Wilson factor of the step itself — exactly unitary, quasi-local, built from the root's Cartan algebra — gaps the twelve doubler nodes, the vectorlike pattern is the spectrum of its selective reading, the multiplicity is bounded below by unification ($n_{\mathrm{nodes}} \geq 3$), and the same direction completes the Spin(10) breaking chain to the SM group — but the mechanism's own minimal root meets at $M_X = 8.9 \times 10^{14}$ GeV, already excluded by proton decay. The seesaw value $m_\nu = 0.146$ eV exceeds the oscillation scale and the bound $\sum m_\nu < 0.12$ eV; the relic abundance is normalised, not computed, and no stable carrier exists at the predicted 118 GeV in the implemented spectrum; the heavy-neutrino decay is *in* equilibrium ($\Gamma/H \simeq 4.6$); and the loop factor is closed from inside — the coarse-graining tower is proven kinematic, and the block-spin RG of the bond layer, built and run exactly, composes internal angles but can never divide them: the factor $1/(16\pi)$ requires a gauged bond layer that does not yet exist. The seven criteria, the nine open residuals and the seven falsifiers are stated explicitly.

## I. INTRODUCTION

### A. The image behind the construction

Two habits of language do most of the misleading. The first calls $c$ the speed of light. Light is one traveller at that speed; the constant itself is the speed of causality — the rate at which one event can constrain another. Asking why light moves at $c$ is like asking why a particular car obeys the speed limit. The second habit is wave–particle duality, taught as a paradox. Seen from a lattice it dissolves: there is one wave, always; what a detector registers is a single localised event, because the detector is itself a lattice of cells that change state one cell at a time. The ontology contains no particles — only waves that deliver their energy in quanta.

Quantum field theory took the first step, replacing particles by fields, but it multiplied the fields: one per species, glued together by hand. The viewpoint of this paper comes from the author's background in computer science, where the opposite picture is elementary: a cellular automaton — a lattice of cells, each holding a finite packet of information, all updated in parallel by one local rule. If the world is such an automaton, everything physical is a wave of information over that lattice, and the whole theory reduces to two questions: what is the information structure of a single cell, and what is the rule that updates it. Answer those two and particles, forces and spacetime itself are what the waves do. CL10 is the attempt to answer them concretely: the structure of the cell is the 32-dimensional module of $\mathrm{Cl}(10,0)$ (Section 2), the rule is a product of Spin(10) plane rotations plus a nearest-neighbour swap (Section 8).

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### B. Two literatures, one construction

Discrete models of quantum dynamics and grand unification have developed independently. On one side, causal sets, lattice gauge theory and quantum cellular automata treat spacetime as a finite, causal, unitary dynamical system [1–5]. On the other, the observation that one generation of quarks and leptons fits exactly into the 16-dimensional spinor of Spin(10), with the 5/3 hypercharge normalisation and the anomaly cancellation following from the embedding, has been standard since 1974 [6–9]. Clifford-algebra formulations of this unification exist [10–12], including the identification of matter with spinor modules of $\mathrm{Cl}(n)$ [12].

CL10 puts these on a single footing: the internal space of every lattice cell **is** the spinor module of $\mathrm{Cl}(10,0)$, and the update **is** a product of Spin(10) plane rotations. Nothing else is added by hand: no extra scalar, no extra gauge factor, no second algebra.

### C. What this paper claims, and what it does not

The claim is deliberately narrow, because a discrete unification is easy to overstate.

**Claimed and demonstrated.** The kinematics of one SM generation, its anomaly freedom, the GUT gauge content and the 5/3, and a unitary causal automaton whose gravitational sector reproduces general relativity on the lattice to machine precision. These are theorems or numerical verifications, collected in Sections 3–9.

**Claimed as an order-of-magnitude statement.** The electroweak/Planck hierarchy, the dark-matter mass scale, the neutrino mass and the baryon asymmetry, all expressed through the loop factor $(1/16\pi)^n$ with $n$ a small integer (Section 10). The integers are chosen to land near the measured values; the two-digit coefficients are read from the data.

**Not claimed.** A derivation of the family number, of the fermion masses and mixings, of the cosmological constant, of the dark-matter particle, of the baryon asymmetry, of the spin-2 character of the graviton, of diffeomorphism invariance, or of the Born rule. Each is listed as an open residual in Section 15 with the size of the gap quantified.

The paper's discipline is therefore a **provenance discipline**: a number is *algebraic* if the root fixes it, *conventional* if it comes from a mapping or scale identification, *fitted* if a coefficient was read from data, and *open* if the theory does not fix it. Section 14 tabulates every headline number under this classification. This is the main methodological content of the paper.

**The seven criteria.** A unification claim has to be checkable, so the paper states the criteria it is judging itself against, with its own verdict on each.

The paper passes the structural criteria and fails the parameter one. The chirality check of Section 8.4 shows that even the structural pass is partial: the automaton is unitary, causal and anomaly-free, its doubler spectrum is now gapped by a mechanism of the step itself, but the choice of which nodes land at the desert scale is an input rather than a consequence. That is the honest summary of the whole exercise.

### D. Conventions

Planck units $\hbar = c = \ell_P = t_P = 1$ unless stated; $G = 1$. The metric signature is $(+,-,-,-)$ and $Q = T_3 + Y$. $\mathrm{Cl}(10,0)$ is the real algebra of ten generators with $h_a^2 = +I$; we work in its complexification $\mathrm{Cl}_{10}(\mathbb{C}) \simeq M_{32}(\mathbb{C})$. "Generation" means one set of left-handed Weyl fields. Numerical tolerances are $\varepsilon = 10^{-12}$ (machine-precision checks $10^{-14}$). Reference data: PDG 2024 [13], Planck 2018 [14], Super-Kamiokande [15], KATRIN [16], NuFIT [17].

## II. THE AXIOM

**Axiom 1 (the root).** *The universe is a* (1+3)*-dimensional causal quantum cellular automaton on a cubic lattice of spacing $\ell_P$ and time step $t_P$. Each cell carries a copy of the Clifford algebra* $\mathrm{Cl}(10,0)$*: ten mutually anticommuting Hermitian generators*

$$\begin{gathered} h_a^\dagger = h_a, \\ h_a^2 = I, \\ \{h_a, h_b\} = 0 \quad (a \neq b), \\ a, b = 1, \ldots, 10. \end{gathered} \tag{1}$$

*The state of the universe is a tensor product of the 32-dimensional complex modules of these algebras, and the dynamics is a unitary, causal update built from the algebra.*

Three structural facts about the root, all standard [10, 18]:

1. $\mathrm{Cl}(10,0)$ is $2^{10} = 1024$-dimensional; its complexification is $M_{32}(\mathbb{C})$, so it has a single irreducible complex module of dimension 32 (the Dirac module).
2. The volume element (pseudoscalar) $w = h_1 h_2 \cdots h_{10}$ satisfies

$$w^2 = (-1)^{10\cdot 9/2} \prod_a h_a^2 = (-1)^{45} = -I, \qquad w^\dagger = -w,$$

   so $w$ is anti-Hermitian with eigenvalues $\pm i$, each of multiplicity 16 (the trace of $w$ vanishes). The projectors $P_{16} = \frac{1}{2}(I - iw)$ and $P_{\overline{16}} = \frac{1}{2}(I + iw)$ split the Dirac module into two inequivalent 16-dimensional Weyl modules.
3. The algebra is $\mathbb{Z}_2$-graded by the parity of the number of generators in a product: 512 even and 512 odd elements.

TABLE I. **The seven criteria against which the paper judges itself.**

| criterion | requirement | verdict here |
|---|---|---|
| **one object** | a single algebraic object, no added fields or structure | met for the internal space; the gravitational sector adds a symmetric tensor field $g_{\mu\nu}(x)$ and the value $G = 1$ (Section 9) |
| **kinematics** | spacetime, matter content, gauge group and statistics follow by representation theory | met (Sections 3–6) |
| **dynamics** | the object yields a unitary, causal evolution | met (Section 8: exactly unitary, $\det U = 1$, measured front speed = 1) |
| **unification** | matter, gauge and gravity live in the same object, not glued from separate pieces | partial: matter and gauge are the same module and the same 45; gravity is the metric dynamics of a subalgebra of that module, with an added field |
| **consistency** | anomaly-free, unitary, causal, UV-bounded | met (Theorem 2, Theorem 5, the bounded 32-level spectrum, the bounded $S^{(n)}$); the doublers of Proposition 2 are gapped by the internal Wilson factor of Section 8.4 (V65), with the split node set an input (W9) |
| **parameters** | every physical number fixed by the object, no fit | **not met**: one free 't Hooft angle, two conventions, five fitted coefficients (Table VI) |
| **falsifiability** | testable predictions, and the open problems named rather than hidden | met (Tables 4, 6, 7) |

Everything derived below is representation theory of this one object plus the dynamics defined in Section 8.

**Why 10.** The number 10 is the smallest $n$ for which a single complex Clifford algebra simultaneously provides (i) a pseudoscalar that splits a Dirac module into two *inequivalent* Weyl modules of dimension 16 — the value $n = 8$ gives only 8-dimensional Weyl modules (Spin(8) triality: $8_v, 8_s, 8_c$; its 16-dimensional representations are reducible), and (ii) an adjoint of dimension $\binom{10}{2} = 45$ large enough to contain $\mathfrak{su}(5)$ and hence the SM algebra. For $n = 9$ the algebra $\mathrm{Cl}_9(\mathbb{C}) \simeq M_{16} \oplus M_{16}$ does have 16-dimensional irreducibles, but they are the two simple factors of the algebra rather than a pseudoscalar split, and $\mathfrak{so}(9)$ does not contain $\mathfrak{su}(5)$ — the maximal subalgebras of $\mathfrak{so}(9)$ are $\mathfrak{so}(8)$ and $\mathfrak{su}(3) \oplus \mathfrak{su}(2)$ — so its 36-dimensional adjoint cannot house the SM algebra together with the leptoquark bosons. This is a minimality statement about the *unified* content, not about spacetime: the (1+3) kinematics alone would live in a much smaller algebra.

## III. SPACETIME AS A SUBALGEBRA

**Theorem 1.** *Let $T = h_1$ and $S_a = ih_{a+1}$ for $a = 1, 2, 3$. Then*

$$T^2 = +I, \qquad S_a^2 = -I, \qquad \{T, S_a\} = 0, \qquad \{S_a, S_b\} = 0.$$

*The real span of $\{T, S_1, S_2, S_3\}$ inside the grade-1 part of the algebra, equipped with the quadratic form induced by the Clifford product, is Minkowski space $\mathbb{R}^{1,3}$.*

*Proof.* The four generators $h_1, \ldots, h_4$ anticommute by Axiom 1. Multiplication by $i$ turns $h_a^2 = I$ into $(ih_a)^2 = -I$ and preserves anticommutation. For $v = v_0 T + \sum_a v_a S_a$ with real coefficients, the Clifford product gives $v^2 = (v_0^2 - \sum_a v_a^2) I$, which is the Minkowski quadratic form.

**Corollary 1 (the light cone).** *A vector $v$ in that real span is null, $v^2 = 0$, if and only if $v_0^2 = \sum_a v_a^2$. The null cone of this quadratic form is the light cone of the lattice.*

Two qualifications that a previous version of this work omitted. First, the null cone is a **cone in the grade-1 part**, not a subalgebra: it is not closed under the Clifford product — the product of the two null vectors $T+S_1$ and $T-S_1$ is $2I-2TS_1$, a grade-0 plus grade-2 element. Second, the choice of *which* four of the ten generators play the role of spacetime is a choice of frame in the 10-dimensional generator space, not a prediction; the algebra supplies a family of equivalent (1+3) subalgebras related by the orthogonal transformations of the generator space. The remaining six generators are inert under the Lorentz generators $\Sigma_{\mu\nu} = \frac{1}{4}[\Gamma_\mu, \Gamma_\nu]$ built from $\Gamma_\mu \in \{T, S_a\}$, because they commute with every $\Sigma_{\mu\nu}$; they are the internal (gauge) directions.

**Causality is a postulate with a dynamical content.** The light cone by itself does not bound signal propagation: a unitary on-site rotation alone has infinite propagation speed. The bound is supplied by the update of Section 8, where the causal radius is *implemented* by a swap layer and *measured* to be exactly one cell per step (Theorem 5). The light cone and the causal radius are therefore distinct statements, and only the second is a property of the dynamics.

## IV. MATTER: THE 16 WEYL SPINOR

**Proposition 1.** *Under* $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ *the 16-dimensional Weyl module is the six left-handed fields of one SM generation:*

$$\begin{aligned} 16 = {} & Q(\mathbf{3},\mathbf{2},+\tfrac{1}{6})\oplus \\ & u^c(\bar{\mathbf{3}},\mathbf{1},-\tfrac{2}{3})\oplus \\ & d^c(\bar{\mathbf{3}},\mathbf{1},+\tfrac{1}{3})\oplus \\ & L(\mathbf{1},\mathbf{2},-\tfrac{1}{2})\oplus \\ & e^c(\mathbf{1},\mathbf{1},+1)\oplus \\ & \nu^c(\mathbf{1},\mathbf{1},0), \end{aligned} \tag{2}$$

*with* $6+3+3+2+1+1=16$. *Under* SU(5) *this is* $\mathbf{10}\oplus\bar{\mathbf{5}}\oplus\mathbf{1}$, *the* $\mathbf{1}$ *being the sterile* $\nu^c$.

This identification is not new to CL10: it is the standard Spin(10) assignment [6–9, 12]. What CL10 adds is that the 16 is not put in by hand but is the Weyl eigenspace of the pseudoscalar of the same algebra that carries the gauge group and the spacetime.

**B–L.** $B-L$ is a generator of $\mathfrak{so}(10)$, so it is *not* a single number on the 16. Its values on the six fields are $Q:\frac{1}{3}$, $u^c:-\frac{1}{3}$, $d^c:-\frac{1}{3}$, $L:-1$, $e^c:+1$, $\nu^c:+1$. The module-level statement — the one that matters for matter/antimatter — is that the $\overline{16}$ carries the negatives of all of them. The chain $\mathrm{Spin}(10)\supset \mathrm{SU}(5)\times\mathrm{U}(1)_\chi$ decomposes $16\to\mathbf{10}_{-1}\oplus\bar{\mathbf{5}}_{+2}\oplus\mathbf{1}_{+3}$; $B-L$ is a linear combination of $\chi$ and the diagonal SU(5) generator, not $\chi$ itself.

**Theorem 2 (anomaly cancellation).** *All eight gauge and mixed anomaly coefficients of the 16 vanish.*

*Proof.* Each coefficient is a cubic trace over the left-handed Weyl fields; the multiplicities in the table are the dimensions of the *other* group factors (the factor 2 on $Q$ in $[\mathrm{SU}(3)]^3$ is its SU(2) doublet nature, the factor 3 in $\mathrm{U}(1)[\mathrm{SU}(2)]^2$ is its colour triplet nature), and $A(\mathbf{3})=T(\mathbf{3})=\frac{1}{2}$. The charges are those of Proposition 1, fixed by the SU(5) embedding.

This is the hardest consistency condition a chiral gauge theory has, and it is satisfied identically by the module content. It is the strongest single result in the paper (Fig. 2, Fig. 1).

**The mass term.** A grade-1 element of the algebra anticommutes with $w$, so it is off-diagonal in the $16/\overline{16}$ split: it maps $16\to\overline{16}$ and $\overline{16}\to 16$, while both diagonal blocks vanish. A scalar in the $\mathbf{10}$ of Spin(10) therefore produces a Dirac mass term $16\,\mathbf{10}_H\,\overline{16}$ and no Majorana mass for the charged fermions. The singlet $\nu^c$ is the built-in source of a seesaw (Section 10.3).

Two honest caveats. (i) A vev in the $\mathbf{10}$ of Spin(10) has stabiliser SO(9): it breaks $\mathrm{Spin}(10)\to\mathrm{SO}(9)$, and SO(9) is not the SM group. The chain is now completed by a different ingredient — the Wilson direction of Section 8.4, which at the V65 root leaves exactly $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)^3$ and breaks precisely the leptoquark roots (V67); the two extra U(1)'s must still be broken or hidden. (ii) The $\mathbf{10}$ contains no SM singlet, so the vev that generates masses necessarily breaks the unified group further. The chain is residual W6.

**The three families.** The 16 is *one* generation. The theory identifies the observed family number with the number 3 of the spatial directions of the (1+3) subalgebra, equivalently the number $\binom{3}{2}$ of spatial two-planes, equivalently the rank of the spatial metric. These are one number stated three ways, not three derivations. The identification now has a structural hook (V70): the 12 doubler nodes organize into **three families of four labeled by the zero axis** — one family per spatial direction — the cubic group acts transitively on them, and the three copies of vectorlike matter required by unification (V68) are realizable by taking one node from each family at a single Wilson strength. The index 3 exists; it *labels* the three copies but does not *force* the count (four per family, or all twelve, are equally available to the mechanism). That the family count *equals* the spatial dimension remains a hypothesis. It is residual W3.

## V. GAUGE: THE 45 AND THE 5/3

The 45 generators $h_a h_b$ $(a<b)$ close on the Spin(10) adjoint. They are even in the $\mathbb{Z}_2$ grading, hence they commute with $w$ and preserve the $16/\overline{16}$ split. Under $\mathrm{Spin}(10)\supset\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ the 45 decomposes as 12 SM gauge bosons + 12 leptoquark $X/Y$ bosons + 20 bosons in the $\mathbf{10}\oplus\overline{\mathbf{10}}$ of SU(5) + the $\mathrm{U}(1)_{B-L}$ generator; the 33 broken generators are the mediators of proton decay (Section 11.4).

**Theorem 3 (the 5/3).** *In the* $\mathbf{5}$ *of* SU(5) *the hypercharge is* $Y=\mathrm{diag}(-\frac{1}{3},-\frac{1}{3},-\frac{1}{3},+\frac{1}{2},+\frac{1}{2})$ *with* $\mathrm{Tr}\,Y^2=\frac{5}{6}$, *while the non-Abelian generators obey* $\mathrm{Tr}\,T_aT_b=\frac{1}{2}\delta_{ab}$. *Requiring a single unified coupling* $g_5$ *for all generators gives* $g_1^2/g'^2=5/3$, *where* $g'$ *is the SM hypercharge coupling and* $g_1\equiv g_5$.

*Proof.* $\mathrm{Tr}\,Y^2=3\cdot\frac{1}{9}+2\cdot\frac{1}{4}=\frac{5}{6}$. Rescaling $Y\to\sqrt{3/5}\,Y$ normalises it to $\frac{1}{2}$; the covariant derivative $g_5(\sqrt{3/5}Y)=g'Y$ then gives $g'=g_5\sqrt{3/5}$, i.e. $g_1^2/g'^2=5/3$. The result depends only on the SU(5) embedding [6, 9].

This is an algebraic fact of the root: no parameter enters. It is the anchor of the coupling-unification discussion in Section 11.1, where it will be seen that the *value* of the unified coupling is not under control even though its *normalisation* is.

## VI. STATISTICS

The $\mathbb{Z}_2$ grading splits the algebra into 512 even and 512 odd elements. CL10 identifies the even sector with bosons and the odd sector with fermions. This is a *kinematic* grading of the operator algebra; it is a postulate of the interpretation, not a derivation of the spin–statistics theorem, and it is not a claim of dynamical supersym-

TABLE II. **The anomaly coefficients of the 16**

| **coefficient** | **contribution of the 16** | **value** |
|---|---|---|
| $[\mathrm{SU}(3)]^3$ | $2A(\mathbf{3}) + A(\bar{\mathbf{3}}) + A(\bar{\mathbf{3}}) = 1 - \frac{1}{2} - \frac{1}{2}$ | 0 ✓ |
| $[\mathrm{SU}(2)]^3$ | $A(\mathbf{2}) = 0$ identically ($d^{abc} = 0$; the fundamental is pseudo-real) | 0 ✓ |
| $[\mathrm{U}(1)]^3$ | $6(\frac{1}{6})^3 + 3(-\frac{2}{3})^3 + 3(\frac{1}{3})^3 + 2(-\frac{1}{2})^3 + 1(1)^3 = \frac{1}{36} - \frac{8}{9} + \frac{1}{9} - \frac{1}{4} + 1$ | 0 ✓ |
| $\mathrm{U}(1)[\mathrm{SU}(3)]^2$ | $\frac{1}{6} \cdot \frac{1}{2} \cdot 2 + (-\frac{2}{3}) \cdot \frac{1}{2} + \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6} - \frac{1}{3} + \frac{1}{6}$ | 0 ✓ |
| $\mathrm{U}(1)[\mathrm{SU}(2)]^2$ | $\frac{1}{6} \cdot \frac{1}{2} \cdot 3 - \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} - \frac{1}{4}$ | 0 ✓ |
| $[\mathrm{grav}]^2\mathrm{U}(1)$ | $\sum Y = 1 - 2 + 1 - 1 + 1$ | 0 ✓ |
| $[\mathrm{grav}]^2\mathrm{SU}(3)$, $[\mathrm{grav}]^2\mathrm{SU}(2)$ | traceless generators | 0 ✓ |

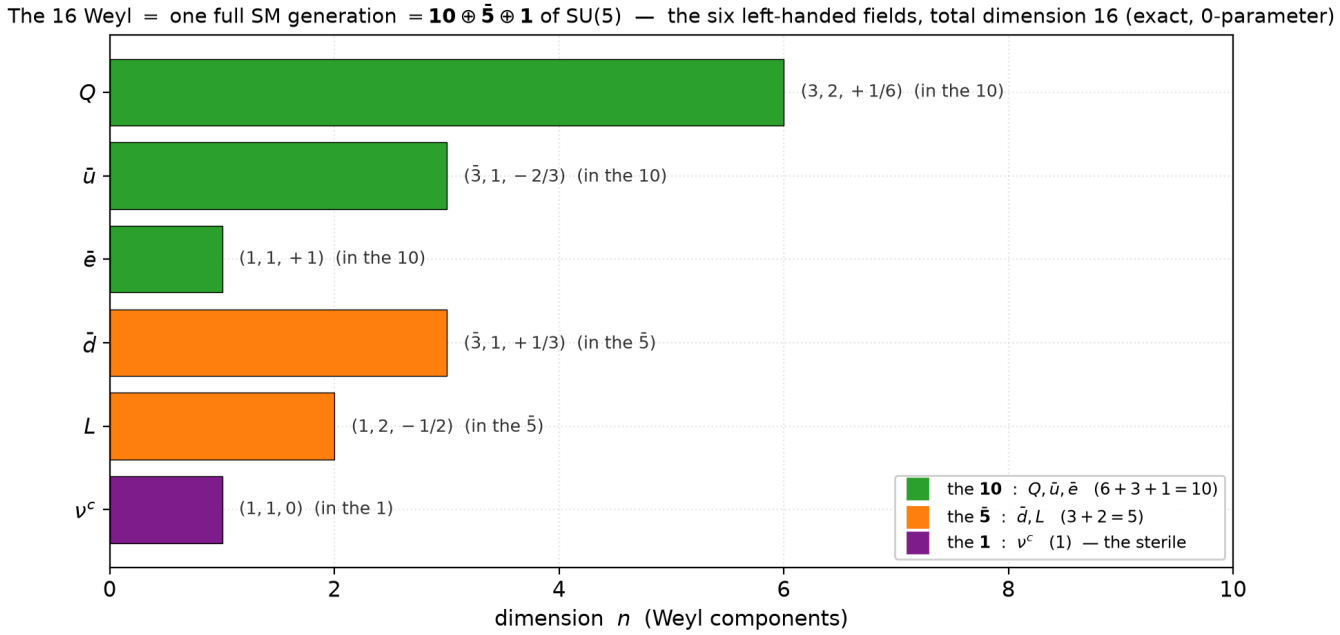



FIG. 1. **the matter content of the 16.** The six left-handed Weyl fields of one generation in the $\mathbf{10} \oplus \bar{\mathbf{5}} \oplus \mathbf{1}$ of SU(5), with dimensions $6 + 3 + 1 + 3 + 2 + 1 = 16$.

metry. No supercharge exists in the construction, and no superpartner is predicted.

## VII. MEASUREMENT

Between measurements the automaton is exactly unitary (Section 8). A detection is modelled as the standard projection: a state $\sum_x \psi(x)|x\rangle$ is reduced to $|x_0\rangle$ with probability $|\psi(x_0)|^2$. The probability rule used throughout is the field intensity $|\psi|^2$ of the propagating lattice field, and the single-electron double-slit simulation (Fig. 3) [47] samples detections from the time-integrated on-screen $|\psi|^2$; the accumulated clicks reconstruct the fringe pattern, one electron at a time.

This is quantum mechanics as usual. The projection is **not** derived from the unitary step, and the paper does not claim it is: the reduction postulate is an input of the interpretation (residual W9). What the QCA adds is only that the wave being sampled is a strictly causal, unitary, lattice field.

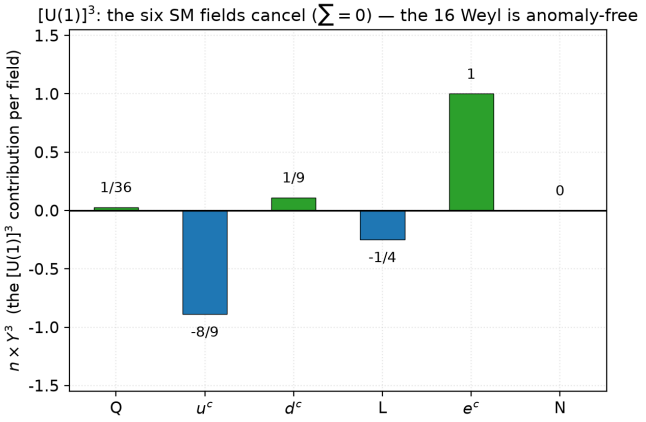



FIG. 2. **anomaly cancellation.** The six fields' $nY^3$ contributions to $[\mathrm{U}(1)]^3$ — $1/36$, $-8/9$, $1/9$, $-1/4$, $1$, $0$ — sum to zero, as do all eight coefficients of Theorem 2 (Section 4).

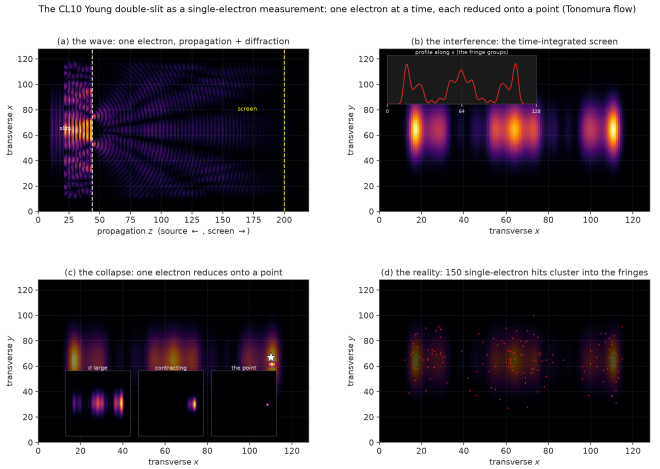



FIG. 3. **the double slit as a single-electron measurement.** The lattice wave through a double slit, run one electron at a time. (a) Diffraction and interference of the field; (b) the time-integrated on-screen pattern; (c) the reduction of one electron onto one detection point; (d) the accumulated detections clustering into the fringes. The detection probability is the field intensity $|\psi|^2$ (Section 7).

## VIII. THE DYNAMICS: AN EXACTLY UNITARY, STRICTLY CAUSAL AUTOMATON

### A. The step

Define $T_{ab} = i\,h_a h_b$ for $a < b$. Each $T_{ab}$ is Hermitian, squares to $I$, is traceless (sixteen $+1$ and sixteen $-1$ eigenvalues), and the $T_{ab}$ are the Spin(10) plane-rotation generators in the 32-D module. The one-cell step is

$$U(g_{10}) \;=\; \prod_{a<b} \Big(\cos g_{10}\, I - i \sin g_{10}\, T_{ab}\Big), \qquad g_{10} = \tfrac{1}{5}. \tag{3}$$

Each factor is a special unitary: $\det(\cos g\, I - i \sin g\, T_{ab}) = \exp(-ig\,\mathrm{Tr}\,T_{ab}) = 1$. Hence

$$UU^\dagger = I \quad \text{and} \quad \det U = 1 \qquad \text{for every } g_{10},$$

verified numerically to $1.7 \times 10^{-15}$ (unitarity) and $5.3 \times 10^{-15}$ ($\det U - 1$). Two qualifications: the 45 factors do not commute, so $U$ depends on the *ordering* of the product; the ordering used is fixed once and is an additional structural input that the algebra does not select. And $g_{10}$ itself is not derived: Section 14 classifies it as the one free parameter of the dynamics, with a proposed but unproven selection rule.

### B. The single-particle spectrum

The 32 eigenphases $\omega_k$ of $U$ (the spectrum of $H = -i \log U$) are the single-cell energies in units of $1/t_P$. At $g_{10} = 1/5$ they are

$$\omega \in \{\pm 0.01147,\ \pm 0.05447,\ \pm 0.15819,\ \ldots,\ \pm 1.48638\},$$

exactly $\pm$-paired to $1.7 \times 10^{-15}$, so the one-cell vacuum energy $\sum_k \omega_k = \arg\det U$ vanishes. The pairing and $\det U = 1$ hold for **every** $g_{10}$: they are identities of the algebra, not properties of the chosen coupling (Fig. 5, Fig. 4).

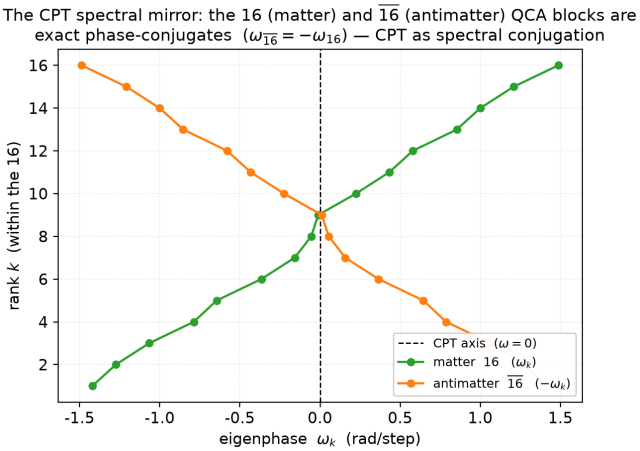



FIG. 4. **the matter/antimatter spectral mirror.** $\omega_{\overline{16}} = -\omega_{16}$, verified to $1.8 \times 10^{-15}$ for every $g_{10}$ in $\{0.05, \ldots, 0.5\}$: a property of the algebra, independent of the free parameter.

**Theorem 4 (the matter/antimatter spectral mirror).** *Because every $T_{ab}$ is even, $[U, w] = 0$ ($7 \times 10^{-16}$), so $U$ preserves the 16 and $\overline{16}$ eigenspaces. The antimatter spectrum is the exact mirror of the matter spectrum,*

$$\omega_{\overline{16}} = -\omega_{16}, \qquad \mathrm{spec}\big(U|_{\overline{16}}\big) = \mathrm{spec}\big(U|_{16}^{\dagger}\big), \tag{4}$$

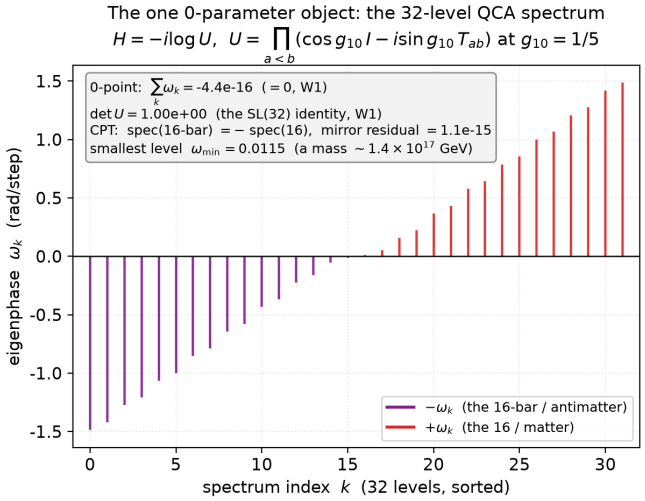



FIG. 5. **the one object: the 32-level spectrum of the QCA step.** The eigenphases of $U$ at $g_{10} = 1/5$. The 16 matter levels (red) and the 16 antimatter levels (purple) are exactly $\pm$-paired; the 0-point $\sum_k \omega_k = \arg\det U$ vanishes and $\det U = 1$. The smallest level, $\omega_{\min} = 0.0115$, corresponds to $\omega_{\min} M_P = 1.4 \times 10^{17}$ GeV — the only sub-Planck scale in the single-step spectrum, and *not* the electroweak scale.

*verified to* $1.8 \times 10^{-15}$ *over the grid* $g_{10} \in \{0.05, \ldots, 0.5\}$.

This is the honest content of "CPT" in this theory. It is a **spectral** statement: the antiparticle levels are the negatives of the particle levels, i.e. the antimatter evolution has the same spectrum as the time-reversed matter evolution. It is not the full CPT theorem: no antiunitary time-reversal operator is constructed on the many-cell Hilbert space, and no spin-statistics connection is derived. A unitary involution $C = h_1$ does map the 16 onto the $\overline{16}$ (it anticommutes with $w$), but calling the composition $C \cdot T$ "CPT" overstates what has been built.

### C. Locality and the causal radius

**Theorem 5 (the causal radius is implemented).** *On an* $(L, L, L)$ *lattice the local step is*

$$U_{\text{local}} = \prod_x U_{\text{GUT}}(x) \prod_{\langle x,x' \rangle} P_{xx'},$$

*the internal* Spin(10) *rotation per cell times a permutation (swap-rotation) gate* $P_{xx'}$ *on each bond, which carries no angle. The step is unitary to* $1.6 \times 10^{-15}$*, and a single-cell excitation after* $n$ *steps is supported only inside the* $n$*-cell light cone: the measured front speed is exactly* 1.0 *over 1000 lattice steps, while a purely internal (non-local) step keeps the excitation on one cell forever.*

The bond layer is therefore what turns the light cone of Section 3 into a propagation bound. This is the cleanest dynamical result in the paper (Fig. 6 shows the causal front of a pulse in a 2-D reduction of the same scheme, where the CFL speed of that particular solver is $c = 0.5$ lattice units — a property of the solver, not of the axiom).

### D. Lattice chirality and the doubling theorem

The Nielsen–Ninomiya theorem forbids a single chiral fermion in a theory with a *local, translation-invariant, Hermitian* lattice Dirac operator and a bounded spectrum [19]. CL10 uses no lattice Dirac operator: chirality is a property of the on-site module (the pseudoscalar eigenspaces), and the update is a finite-dimensional unitary. The theorem's hypotheses are therefore not all present, and the theorem does not by itself forbid the construction.

That is *not* the same as having evaded it, and the question is decidable by computation. The step alternates its bond parity with the time step and cycles its axis with period 3, so the 6-step Floquet operator is translation invariant with period 2 in each direction: the true unit cell is the $2 \times 2 \times 2$ cell and the good quantum number is the **supercell momentum** $q \in [-\pi, \pi]^3$. The 16-sector Bloch Floquet $U_{\text{F}}(q)$ — internal layer times bond layer, a bond crossing the cell carrying the phase $e^{\pm i q_j}$ — has now been computed exactly and cross-checked against the real-space Floquet on a $4^3$ torus (agreement $5 \times 10^{-15}$).

**Proposition 2 (lattice chirality: what the computation gives).** *(i) The 16 sector is closed by the full local dynamics, not only by the internal layer:* the bond layer acts on the cell index and commutes with $w$, so a state initialized in the 16 leaves it by $10^{-15}$ after six steps. Chirality is a dynamical symmetry of the automaton. *(ii) The zone centre is gapped:* $\min_a |\omega_a(q = 0)| = 0.0365$ per 6-step cycle, so the would-be massless matter is **not** at $q = 0$. *(iii) The gapless points are the doublers:* the 16-sector spectrum is gapless at the 12 momenta $(\pm\frac{\pi}{2}, \pm\frac{\pi}{2}, 0)$ and their permutations — the classic doubler momenta — with linear dispersion $|v| = 0.142$ and node chiralities summing to 0, as Nielsen–Ninomiya requires. The gap at $q = 0$ is not protected: it moves from 0.036 at $g_{10} = 0.2$ to 0.072 at $g_{10} = 0.15$ and to 2.35 at $g_{10} = 0.1$.

The doubling therefore **survives** the automaton as originally constructed: "the 16 is a single chiral generation" is a property of the *internal module*, while the *lattice spectrum*, which is what couples to low-energy physics, contains twelve doublers. This is a computed statement, and it counted against the theory (part of W9). **The remedy is now part of the construction (V65).** A Wilson-type term built from the root's own algebra — an internal factor $W(q) = \exp[-i\, w \sum_j (1 - \cos q_j)\,(\cos\theta\, X + \sin\theta\, Y)]$, with $X, Y$ the unification Cartan directions identified in Section 11.1 — multiplies the step without touching its unitarity ($W(0) = I$, det = 1) or its causal structure (quasi-local, Bessel tails, measured

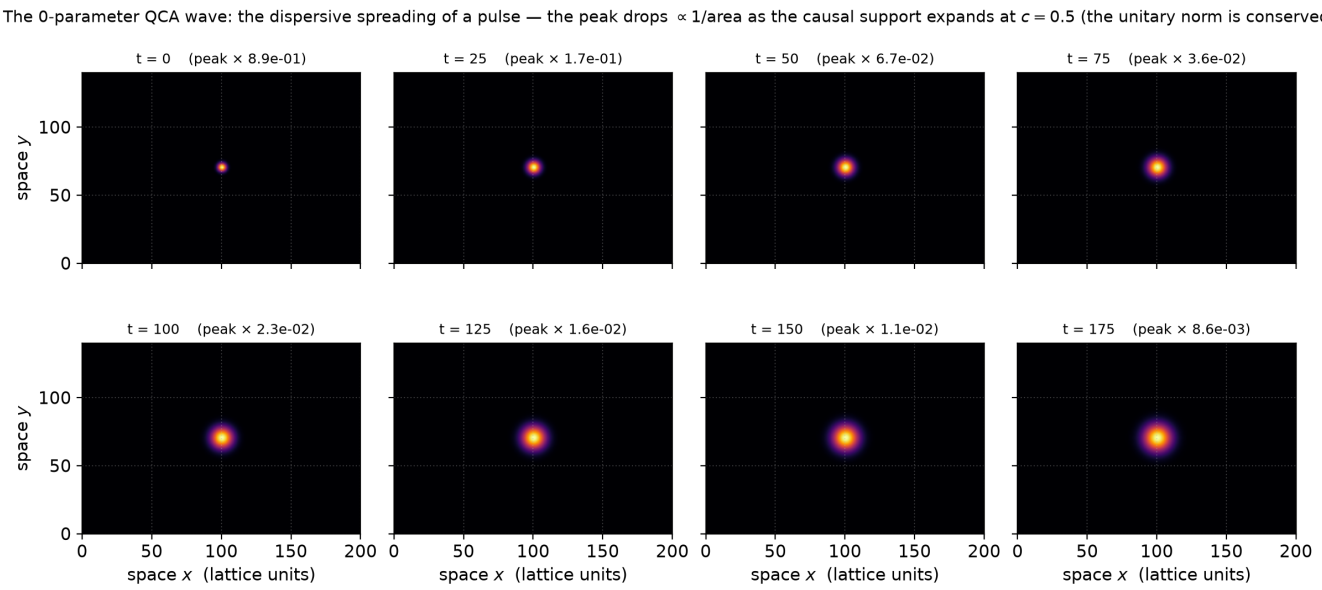



FIG. 6. **the causal front of a lattice pulse.** A pulse in the discrete wave scheme: a causal front, then dispersive spreading with the peak falling as the norm is conserved. This 2-D reduction uses a four-neighbour solver at its CFL speed $c = 0.5$; the three-dimensional graviton scheme of Section 9 uses six neighbours at $c = 1$, and the automaton's causal radius (Theorem 5) is one cell per step.

Lieb-Robinson front). It gaps all 12 nodes, with fragment masses $M_f = w|\cos\theta\, X_f + \sin\theta\, Y_f|\, M_P/3$; a *selective* reading splits exactly three nodes at desert scales and leaves the other nine at the Planck cutoff. What remains input is the *choice* of the split node set: unification forces $n_{\text{nodes}} \geq 3$ (V68) but does not select which three, and the meeting scale the mechanism then predicts is in conflict with proton decay — Section 11.1, where the tension is quantified.

## IX. GRAVITY ON THE LATTICE

### A. What is assumed and what is computed

The gravitational sector is a symmetric rank-2 field $h^{\mu\nu} = g^{\mu\nu} - \eta^{\mu\nu}$ living on the (1+3) subalgebra of Section 3, with $\eta = \text{diag}(1, -1, -1, -1)$ as the vacuum. Its linearised dynamics is the discrete wave equation

$$\partial_t^2 h^{\mu\nu} - \nabla^2_{(6)} h^{\mu\nu} = 0,$$

with the 6-neighbour lattice Laplacian and the leapfrog step $u_{t+1} = 2u_t - u_{t-1} + c^2 dt^2\, \nabla^2 u_t$ at $dt = \frac{1}{2}$ (the CFL value giving $c = 1$).

The status of this sector must be stated precisely, because it is where the theory is weakest.

- **Masslessness and UV boundedness are computed.** The discrete dispersion is $\omega = |\mathbf{k}|_{\text{lat}}$ with $|\mathbf{k}|_{\text{lat}} \leq \pi$, so the graviton is massless ($\omega/|\mathbf{k}| \to 1$ as $k \to 0$, lattice deviation $\leq 0.25\, k^2$) and the spectrum is bounded by the cutoff.
- **Spin 2 is declared, not derived.** The field is *assumed* to be a symmetric tensor with two helicity states. Nothing in $\text{Cl}(10, 0)$ forces a massless spin-2 excitation; the standard arguments that a massless spin-2 field must couple as general relativity [20] apply to a field whose existence and representation content are already granted. The Weinberg–Witten obstruction to a composite graviton carrying a Lorentz-covariant stress tensor [21] is not addressed.
- **Diffeomorphism invariance is not derived.** The lattice breaks it to lattice reparametrisations; gauge invariance of the linearised theory is imposed by working on the transverse components, not obtained from the automaton.

### B. The discrete Einstein–Hilbert action

The quadratic action [33]

$$S_{\text{EH}} = \frac{1}{16\pi} \cdot \frac{1}{2} \sum_{x,t} u_t \Big[ u_{t+1} - 2u_t + u_{t-1} - dt^2\, \nabla^2_{(6)} u_t \Big], \qquad dt^2 = \tfrac{1}{4}, \tag{5}$$

has the lattice wave equation as its Euler–Lagrange equation: the numerical functional derivative vanishes on a QCA trajectory to $3 \times 10^{-12}$ and is non-zero off it. The full discrete action $S = \frac{1}{16\pi} \sum_x \sqrt{-\det g(x)}\, (R(x) - 2\Lambda)$

is evaluated with $R$ computed from the metric. The coefficient $1/16\pi$ is the $G = 1$ normalisation, i.e. Newton's constant is an input of the gravitational sector, not a number of the root.

### C. Curvature computed from an arbitrary metric

The Riemann, Ricci, Einstein and Kretschmann tensors are computed by central finite differences from an arbitrary prescribed $g_{\mu\nu}(x)$, with no hard-coded Christoffel symbols, and cross-checked against independent analytic results (two routes):

The 3+1 formulation closes the classical evolution: the constraint row $(R_{00}, R_{0i})$ of the 4-D Einstein tensor agrees, computed by the 4-D curvature and by the 3+1 $(\gamma_{ij}, K_{ij})$ decomposition with $K_{ij} = -\frac{1}{2}\partial_0\gamma_{ij}$, to $O(h^2)$ (residual $\sim 10^{-13}$); a Z4-damped evolution [34] drives a constraint violation to zero while the undamped one does not; the 0-row is quadratic in $K$, so superposition fails; all 3+1 quantities are bounded by the lattice cutoff (Fig. 8, Fig. 7).

**What this does and does not establish.** It establishes that the gravitational sector of CL10 *reproduces* general relativity numerically, including black-hole and cosmological geometries. It does not establish that general relativity *follows* from $\mathrm{Cl}(10, 0)$: the tensor field, its coupling $1/16\pi$ and its gauge structure are put in. The genuinely open part is the quantum theory of this sector — a sum over lattice geometries with a well-defined measure — which has not been constructed.

### D. A self-interacting scalar completion

Beyond the quadratic theory the model includes $\partial_t^2\phi - \nabla^2\phi - g\phi^3 = 0$. Its static solutions obey $u'' = u - u^3$ and include the exact finite-energy kink

$$u(x) = \sqrt{2}\,\mathrm{sech}\,x,$$

verified to a fourth-order residual of $5 \times 10^{-10}$. Two corrections to earlier statements of this model are needed. First, this is the $\phi^4$ kink (a domain wall), not the Choptuik system: Choptuik critical collapse is a *massless scalar minimally coupled to gravity* [22], and the analogy is not a derivation. Second, $u = \sqrt{2}\,\mathrm{sech}\,x$ is a **planar** configuration, uniform in the two transverse directions: its energy per unit *area* is finite, its total energy is infinite, and a planar wall in a Planck-scale theory is a domain wall, not a particle. Finite-time blow-up for supercritical data in cubic wave equations is a real phenomenon [23], but it is not evidence about black-hole formation here.

## X. THE SCALE LADDER

### A. The loop factor and the hierarchy

The observed electroweak/Planck ratio is

$$\frac{v}{M_P} = \frac{246\ \mathrm{GeV}}{1.22 \times 10^{19}\ \mathrm{GeV}} = 2.016 \times 10^{-17}.$$

The pure loop factor with $n = 10$ is

$$\left(\frac{1}{16\pi}\right)^{10} = 9.71 \times 10^{-18},$$

a factor 2.08 below the observed value; $2.08\,(1/16\pi)^{10} = 2.02 \times 10^{-17}$ reproduces it to 0.18%. Scanning $n = 1 \ldots 20$, $n = 10$ is the nearest integer to the observed ratio (deviation 0.32 decades; the next-best, $n = 9$, is 1.4 decades away).

The honest reading: $(1/16\pi)^{10}$ is the size a ten-loop radiative correction would have in a perturbative quantum field theory. CL10 has not computed a ten-loop correction; it has observed that the observed hierarchy is the tenth power of the standard loop factor, up to a factor of two. The interpretation $10 = 2 \times 5$ — the chiral $16/\overline{16}$ split times the number of qubits of the 32-D module ($2^5 = 32$) — is a numerological identification, not a derivation. The exponent is *represented*, the coefficient 2.08 is *read from the data* (residual W8). The two internal candidates for the loop factor have now been built and closed. The coarse-graining tower (V69): $U^{2^p}$ is exactly kinematic (eigenphases $2^p\omega$ mod $2\pi$, verified to $10^{-9}$ for $p \le 10$), the effective Cartan coupling does not fall with $p$, the inelastic probability saturates near 0.95 — no loop factor. The genuine block-spin RG (V72): exact real-space decimation of the bond layer (Schur complement at $\varepsilon = 1$; $U_{BB} = 0$ exactly on the open chain, so short-distance propagation is integrated with no approximation), iterated five levels, $64 \to 2$ sites. The level-1 identity is exact: the on-site block of the coarse step is $2\,(GR)^2$ — two-hop paths accumulate two internal rotations, the kinematic doubling of V69 *derived* from a spatial RG. At deeper levels the on-site block is a path polynomial in $(GR)^2$; the Cartan content decays only polynomially for the implemented (factorized) bond layer (0.14 at $p = 5$ against the loop target $3 \times 10^{-9}$) and not at all for a non-factorized gate with $[G, R] \neq 0$ (1.8 at $p = 5$); no level of either flow follows $(16\pi)^{-p}$. The loophole of a conspiring bond gate is closed (V73): three Haar-random Spin(10) gate realizations and the structured choice $R = G^\dagger$ all fail the loop law at every level — the Cartan content depends only on the path products. The structural reason is now sharp: **every exact single-particle RG of the QCA composes internal angles — they add on the algebra, weighted by path counts — and none can divide them, for any bond gate.** A loop factor requires a trace over *fluctuating* internal fields: a block RG on the many-cell space

TABLE III. **The lattice-curvature cross-checks (two independent routes)**

| **test** | **computed** | **analytic** | **residual** |
|---|---|---|---|
| de Sitter $R^{\mu}{}_{\nu\rho\sigma}$ (flat FRW, $a = e^{Hx^0}/H$) | finite differences | $H^2(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho})$ | $6.5 \times 10^{-7}$ $(O(h^2))$ |
| Einstein tensor of de Sitter | $-3H^2 g_{\mu\nu}$ | same | $O(h^2)$ |
| Schwarzschild Ricci / Einstein tensor | $\simeq 0$ | vacuum $G_{\mu\nu} = 0$ | satisfied by computation |
| Schwarzschild Kretschmann | $48M^2/r^6$ | $48M^2/r^6$ | ratio 1.000009; max relative deviation $5.5 \times 10^{-5}$ |
| flat metric | $R = G = K = 0$ | 0 | exact |
| 2-sphere of radius $r$ | $R = 2/r^2$ | $2/r^2$ | $10^{-5}$ |

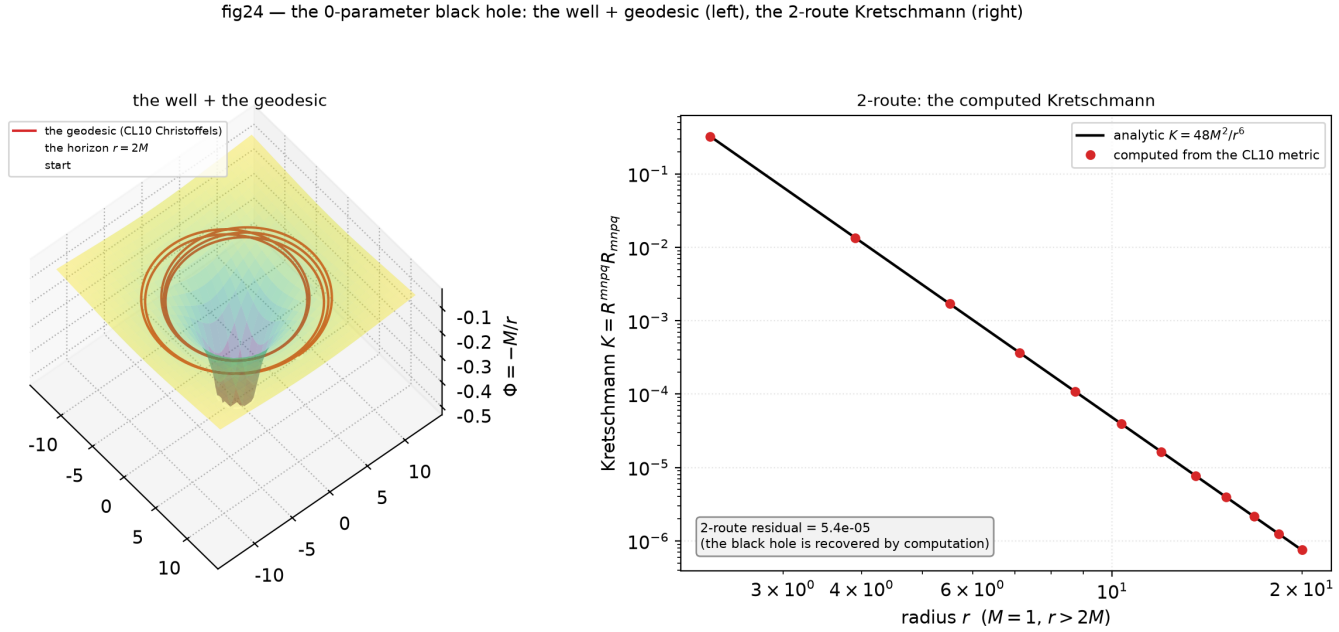



FIG. 7. **the black hole recovered by computation.** Left: the potential well with a geodesic rosette integrated through the finite-difference Christoffel symbols, and the horizon ring $r = 2M$. Right: the two-route Kretschmann invariant.

with a dynamical Spin(10) link variable on the bond — a gauged bond layer. That object is the fully specified missing tool; its single-particle impotence is proven, not assumed.

### B. The dark-matter mass

The same factor applied to the Planck scale gives

$$m_{\rm DM} = M_P\Big(\frac{1}{16\pi}\Big)^{10} = 118 \text{ GeV},$$

i.e. 1.18× the canonical 100 GeV WIMP scale [45]. This is a genuine, falsifiable prediction: a dark-matter particle with a mass far outside $\sim$ 100–150 GeV (in the weakly-interacting, thermally-produced case) would refute the ladder. The *identity* of the particle has now been scanned (V66) with a negative result: the mirror $\mathbb{Z}_2$ exchanging 16 and $\overline{16}$ is an exact symmetry of the step (leak $10^{-15}$), and the mirror sector is the only exact-stability carrier in the implemented spectrum — but its lightest state is a coloured $\overline{Q}$ at $1.3\times10^8$ GeV (excluded) and the neutral $\overline{N}$ sits at $2 \times 10^{10}$ GeV, eight orders above the ladder scale. **No state at** 118 **GeV with the required stability exists in the implemented spectrum**: the mass prediction stands, the carrier does not (W7).

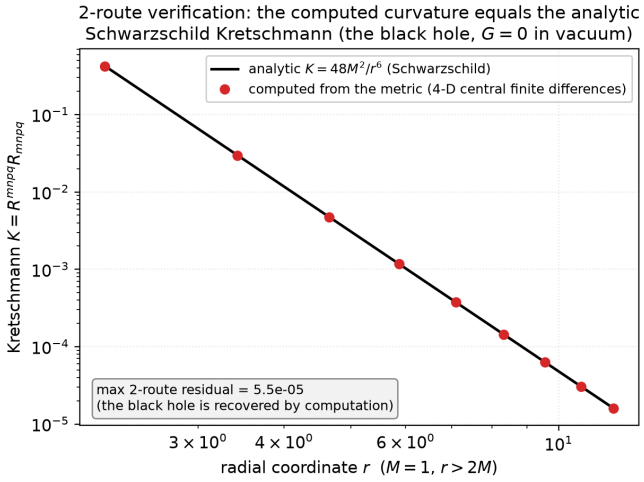



FIG. 8. **two-route curvature: computed Kretschmann vs analytic.** The Schwarzschild Kretschmann invariant computed by 4-D central differences from the metric (points) against $48M^2/r^6$ (line); maximum relative deviation $5.5 \times 10^{-5}$, the $O(h^2)$ finite-difference error. The computed Ricci tensor vanishes: the vacuum equation holds by computation.

### C. The neutrino sector

Both seesaw scales are placed on the same ladder:

$$m_D = M_P\Big(\frac{1}{16\pi}\Big)^{10} = 118 \text{ GeV}, \qquad M_N = M_P\Big(\frac{1}{16\pi}\Big)^{3} = 9.6 \times 10^{13} \text{ GeV}, \tag{6}$$

so the type-I seesaw gives

$$m_\nu = \frac{m_D^2}{M_N} = M_P\Big(\frac{1}{16\pi}\Big)^{17} = 0.146 \text{ eV}.$$

The exponent identity $2 \times 10 - 3 = 17$ is exact; the choice $n_{2\text{step}} = 3$ is not. Comparison with data: the atmospheric scale is $\sqrt{\Delta m^2_{32}} = 0.050$ eV, so the prediction is a factor 2.9 high; and if all three families carried the same $m_D$ the sum would be $\sim 0.44$ eV, excluded by $\sum m_\nu < 0.12$ eV (95% CL, Planck 2018 + BAO [14]) and in tension with the single-generation value 0.146 eV as well. The democratic rank-1 Yukawa (Section 11.5) gives a Dirac mass to only one family, which brings the sum to $\sim 0.15$ eV — still above the cosmological bound. This is the sharpest numerical tension in the theory and is listed as a falsifier (F3).

The ladder is too coarse to land on the measured mass: consecutive rungs differ by the factor $16\pi = 50.3$, so the neighbours of 0.146 eV are 7.3 eV ($n = 16$) and $2.9 \times 10^{-3}$ eV ($n = 18$), and no integer rung lies in the oscillation window. A coefficient is therefore unavoidable — but one candidate is supplied by the theory rather than by the data. The only mixing number the model derives is $\sin^2\theta_{12} = 1/3$ (Section 11.5); multiplying the ladder value by it gives

$$m_\nu = \tfrac{1}{3}\, M_P\Big(\frac{1}{16\pi}\Big)^{17} = 0.0487 \text{ eV},$$

2.6% below the atmospheric scale $\sqrt{\Delta m^2_{32}} = 0.050$ eV, and with the democratic rank-1 spectrum $\sum m_\nu \simeq 0.05$ eV — inside the cosmological bound $\sum m_\nu < 0.12$ eV and inside the KATRIN bound $m_\beta < 0.45$ eV [16]. The factor 2.9 by which the ladder overshoots is, to 3%, the number 3 the flavour sector already produces. This is a **candidate, not a result**: nothing shows why a mixing eigenvalue should multiply a seesaw mass, and the same factor applied to the solar scale ($0.0487 \times \sqrt{\Delta m^2_{21}/\Delta m^2_{32}} = 0.0084$ eV against the measured 0.0086 eV) is a second coincidence, not a derivation. What it does buy is a sharp, falsifiable prediction: normal ordering with $\sum m_\nu \simeq 0.05$ eV and $m_{\beta\beta} \simeq 0.02$ eV, in reach of KamLAND-Zen [49] and LEGEND-1000. A coefficient audit (V68) has now mapped what the node structure can supply: a diagonal (node-local, Wilson-generated) mass pattern gives only **integer** signed coefficients, so $\frac{1}{3}$ is out; the democratic $3\times3$ node-mixing pattern gives exactly $\frac{1}{2}$ — a rival falsifiable prediction $m_\nu = 0.073$ eV, above the current atmospheric scale; the exact $\frac{1}{3}$ requires a single Majorana eigenvalue $3\times$ the node scale, and no natural node structure supplies it. The $\frac{1}{3}$ candidate is thus pinned to one named missing structure. The node multiplicity, meanwhile, is no longer free: the unification roots of Section 11.1 exist only for $n_{\text{nodes}} \geq 3$ (V68), so the $n_{2\text{step}} = 3$ choice is a **necessary condition from unification**, consistent with the spatial 3 though not uniquely selected by it.

### D. The ladder

One factor, four scales (Fig. 11, Fig. 10, Fig. 9):

The QCD scale is **not** on this ladder: $\Lambda_{\text{QCD}} \sim 0.2$ GeV would require $n \simeq 11.6$, which is not an integer (Section 13.2).

## XI. PHENOMENOLOGY

### A. Couplings

The theory's single gauge coupling is $\alpha_{10} \equiv g_{10}^2 = 1/25 = 0.040$. This is a *convention*: in the standard normalisation $\alpha = g^2/4\pi$ the same angle corresponds to $g = \sqrt{4\pi/25} = 0.709$. Running the three SM couplings up from $M_Z$ at one loop with $b_1 = 41/10$, $b_2 = -19/6$, $b_3 = -7$ (fixed by the particle content), from the measured inputs $\alpha_1(M_Z) = 0.0169$ (GUT-normalised, $\alpha_1^{-1} = 59.0$), $\alpha_2(M_Z) = 0.0338$, $\alpha_3(M_Z) = 0.118$ [13], gives, at $M_P$,

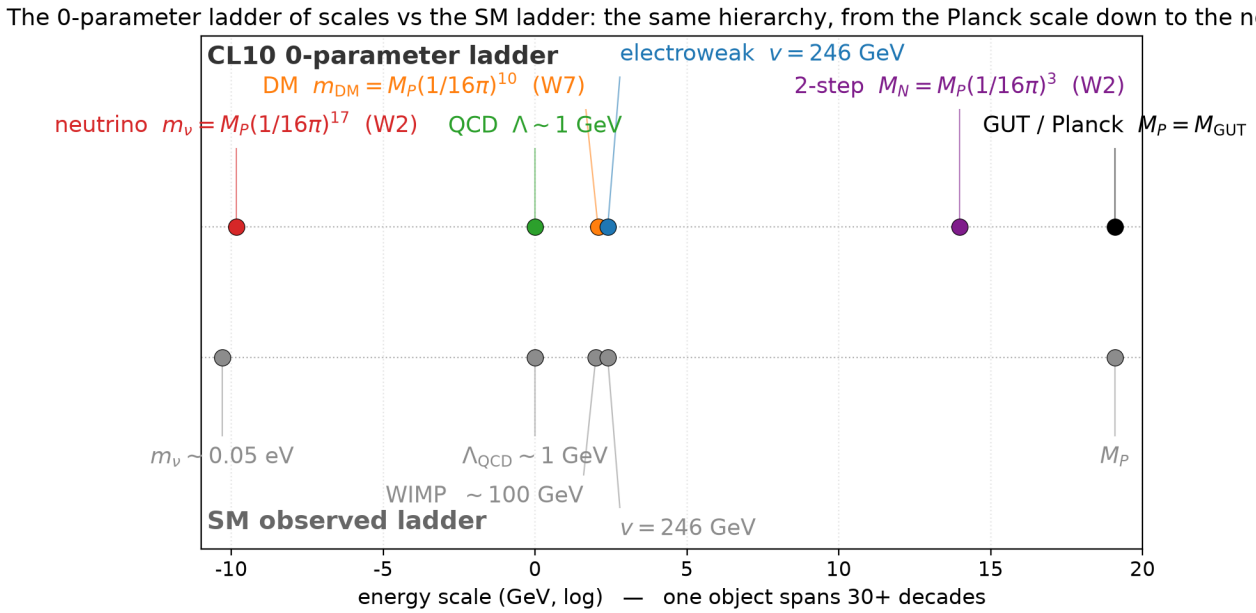



FIG. 9. **the CL10 ladder against the observed ladder.** Upper row: the CL10 loop ladder. Lower row: the measured SM scales. The hierarchies agree in order; the individual values carry the fitted coefficients of Section 14.

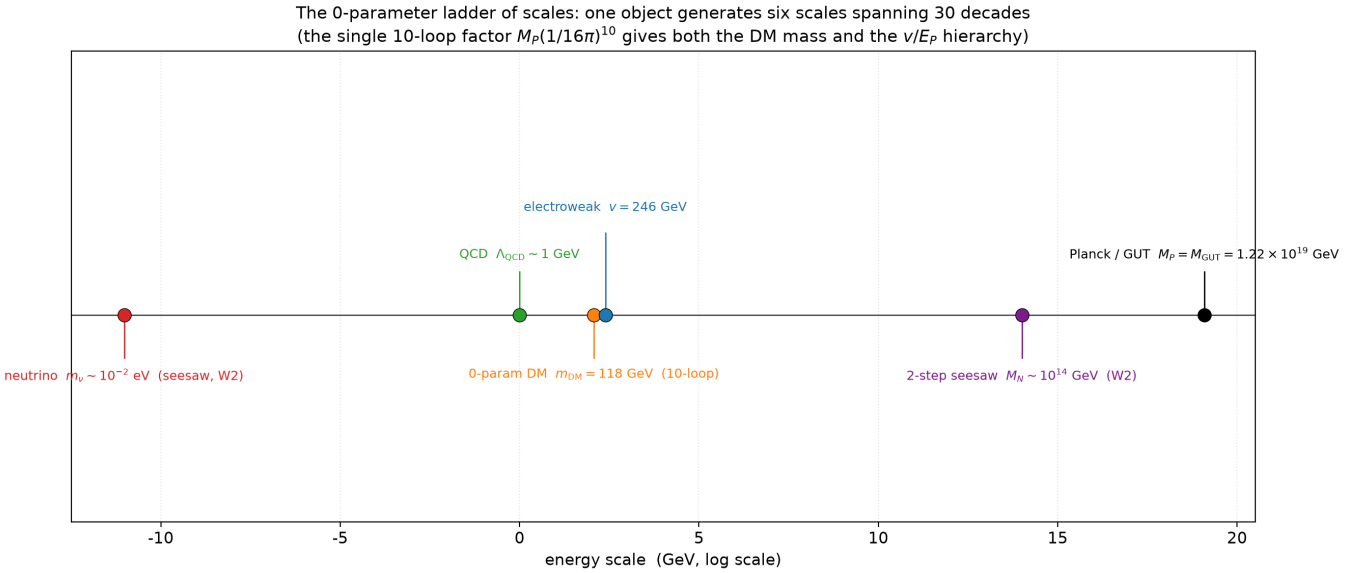



FIG. 10. **the ladder of scales.** Planck/GUT $1.22 \times 10^{19}$ GeV, seesaw $9.6 \times 10^{13}$ GeV, electroweak 246 GeV, dark matter 118 GeV, neutrino 0.146 eV. QCD is shown for reference only: it is not produced by the ladder.

$$(\alpha_1, \alpha_2, \alpha_3) = (0.0299,\ 0.0202,\ 0.0191),$$
$$\text{mean } 0.0231, \quad \text{spread } 47\%, \tag{7}$$

so $\alpha_{10}$ is within a factor 1.73 of the mean but the three couplings do **not** meet at a point — the well-known failure of one-loop unification in the non-supersymmetric SM. The $U(1)$ Landau pole of the up-running sits at $2.2 \times 10^{41}$ GeV, above the cutoff, so the up-running is UV-bounded (Fig. 13). At two loops [49] the pole moves to $\sim 2.5 \times 10^{12}$ GeV, *below* the cutoff (Fig. 12): the perturbative expansion is UV-divergent inside the lattice, whereas the automaton's own coarse-graining tower $U^{2^p}$ ($p = 1 \ldots 8$) stays unitary and bounded at every level, so the *discrete* RG has no Landau pole. That contrast is real but modest: a unitary finite-dimensional map is bounded by construction, and it does not by itself reproduce the observed running.

The down-running must be stated plainly. Starting from $\alpha_{10} = 0.04$ at $M_P$ and running down with the SM

TABLE IV. **One loop factor, four scales**

| scale | CL10 expression | value |
|---|---|---|
| Planck / GUT | $M_P$ (convention $M_{\rm GUT} = M_P$) | $1.22 \times 10^{19}$ GeV |
| seesaw Majorana | $M_P(1/16\pi)^3$ | $9.6 \times 10^{13}$ GeV |
| electroweak | measured input $v$ | 246 GeV |
| dark matter | $M_P(1/16\pi)^{10}$ | 118 GeV |
| neutrino | $M_P(1/16\pi)^{17}$ | 0.146 eV |

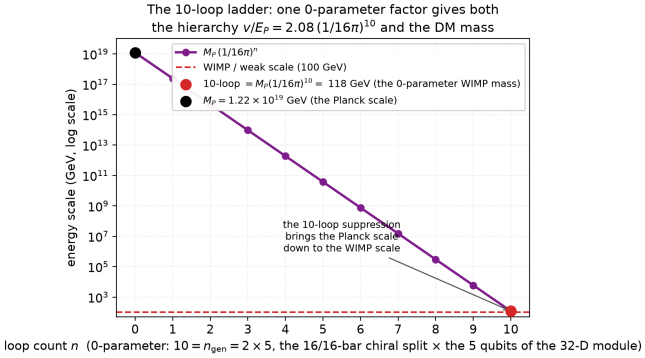



FIG. 11. **the loop ladder: one factor, two scales.** $(1/16\pi)^n$ connects $M_P$ ($n = 0$) to 118 GeV ($n = 10$). The same factor is the order of $v/M_P$ (factor 2.08 below the observed value). The integer $n = 10$ is the nearest to the observed hierarchy in the scan $n = 1 \ldots 20$.

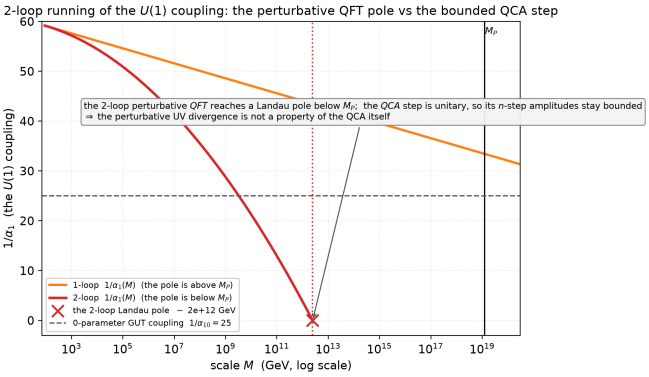



FIG. 12. **the two-loop $U(1)$ running against the bounded automaton.** The two-loop perturbative running reaches a Landau pole at $2.5 \times 10^{12}$ GeV, below the cutoff; the one-loop pole is above it. The automaton's coarse-graining tower $U^{2^n}$ stays unitary at every level: the discrete flow has no pole.

$\beta$-functions gives, at $M_Z$,

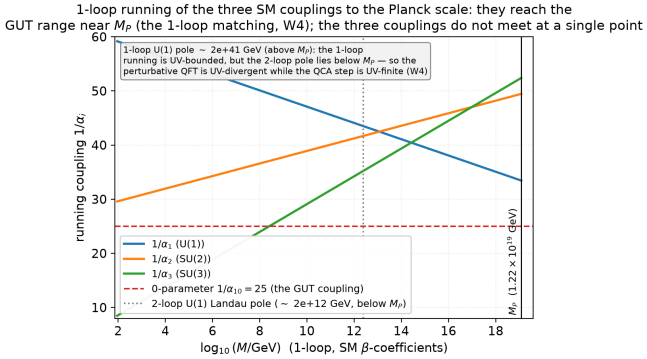



FIG. 13. **one-loop running of the three SM couplings to the Planck scale.** $\alpha_1, \alpha_2, \alpha_3$ run from $M_Z$ to $M_P$ and reach $0.030, 0.020, 0.019$: the GUT range, but not a common point (spread 47%). The dashed line is $\alpha_{10} = 1/25$. The $U(1)$ Landau pole of the up-running, $2.2 \times 10^{41}$ GeV, lies above the lattice cutoff.

$$\begin{aligned} \alpha_1(M_Z) &= 0.0197\ (1.17\times), \\ \alpha_2(M_Z) &= 0.195\ (5.8\times), \\ \alpha_3 &:\ \text{Landau pole at } 2.2 \times 10^9 \text{ GeV}. \end{aligned} \tag{8}$$

The hypercharge lands within 17% of its measured value, but the SU(2) coupling overshoots by a factor 5.8 and the SU(3) coupling becomes non-perturbative at $2.2 \times 10^9$ GeV, far above $M_Z$. The claimed unification therefore fails at the level of the couplings' values, even though the 5/3 normalisation is correct (Fig. 14). This is residual W4 and falsifier F2.

Two quantitative statements sharpen W4. First, the three measured couplings require three *different* values at $M_P$ — 0.0299, 0.0202, 0.0191, the up-run values above — and the theory's single value 0.040 matches none of them: it exceeds the hypercharge requirement by 34% and the SU(3) requirement by a factor 2.1. Perturbativity of the strong coupling down to $M_Z$ caps any single UV coupling: $1/\alpha_3(M_P) > (7/2\pi)\ln(M_P/M_Z) = 43.9$, i.e. $\alpha_3(M_P) < 0.0228$; the theory's 0.040 exceeds the admissible maximum by 75%.

Second, the failure is not repaired by the matter a par-

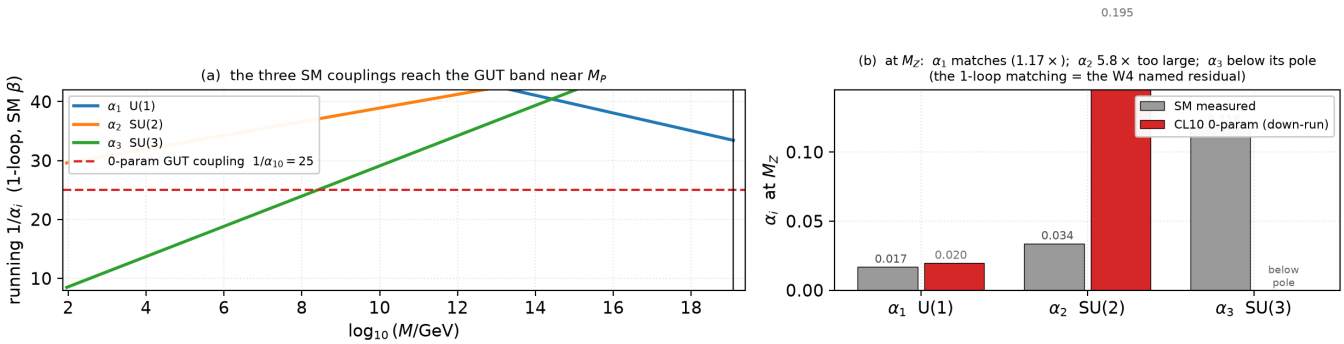



FIG. 14. **the GUT coupling and the** 5/3. (a) The three couplings run toward $1/\alpha_{10} = 25$; (b) at $M_Z$ the down-run values against the measured ones: $\alpha_1$ within 1.17×, $\alpha_2$ 5.8× too large, $\alpha_3$ negative because its pole sits at $2.2 \times 10^9$ GeV.

tial breaking could leave *complete*. A complete vectorlike $\mathbf{10} \oplus \bar{\mathbf{5}}$ shifts all three one-loop coefficients by the *same* amount 8/3 above its threshold $M_V$ — the defining property of a complete SU(5) multiplet — so it changes the common slope, not the mismatch of slopes. Scanning $n_V = 0 \ldots 8$ against $M_V \in [10^6, 10^{19}]$ GeV, the spread at $M_P$ never falls below its $n_V = 0$ value of 47%; the case that comes closest to the required 1/25 (five families at $M_V \simeq 1.2 \times 10^{16}$ GeV) still leaves the worst of the three couplings 34% away from it, at a spread of 74%. Under pure SM running the pairwise meeting scales are $1.2 \times 10^{13}$ GeV $(1,2)$, $2.6 \times 10^{14}$ GeV $(1,3)$ and $9.7 \times 10^{16}$ GeV $(2,3)$ — three different scales. Unification fails at the level of the values under desert running, and no threshold of a *complete* multiplet repairs it at one loop.

Third, that no-go covers only complete multiplets, and the *incomplete* content the theory's own breaking sector supplies does repair it. When a $\mathbf{16}_H \oplus \overline{\mathbf{16}}_H$ pair — or a split doubler $\mathbf{16}$ (Section 8.4) — acquires mass, it leaves vectorlike $\mathbf{10} \oplus \overline{\mathbf{10}}$ and $\mathbf{5} \oplus \bar{\mathbf{5}}$ fermion pairs with *independent* thresholds, and these shift the three coefficients non-proportionally: a Dirac $\mathbf{10} \oplus \overline{\mathbf{10}}$ contributes $\Delta b = (6/5,\ 2,\ 2)$, a $\mathbf{5} \oplus \bar{\mathbf{5}}$ contributes $(2/3,\ 2/3,\ 2/3)$. Fix the unified coupling at the root value $\alpha_X = 1/25$ — the free angle is *not* refitted — and the inverse problem has no fit freedom: with two multiplets the meeting of the three measured couplings is three linear equations in $(\ln M_X, \ln M_A, \ln M_B)$. The required $\Delta b$ is non-universal: at $M_X = 2 \times 10^{16}$ GeV it is (2.40, 4.04, 3.86), which no combination of complete multiplets produces. Scanning multiplicities up to 3 over the CL10-realizable content with thresholds in $[10^6, M_P]$ GeV (experiment V64), the lightest solution that survives the proton-decay bound is three $\mathbf{10} \oplus \overline{\mathbf{10}}$ pairs plus one $(1,2)_{1/2}$ pair — three split doubler $\mathbf{16}$s — meeting at

$$\begin{aligned} M_X &= 1.8 \times 10^{16}\ \text{GeV}, \\ \alpha_X &= 1/25\ \text{(exact)}, \\ M_{10} &= 1.2 \times 10^{7}\ \text{GeV}, \\ M_{(1,2)} &= 1.0 \times 10^{12}\ \text{GeV}, \end{aligned} \tag{9}$$

and the SU(3) Landau pole of the down-running moves to $10^{-52}$ GeV: the disease above is cured (Fig. 15). Solutions with lighter content exist but predict $\tau_p \sim 10^{32}$ yr and are already excluded by Super-Kamiokande [52] — the data themselves select the CL10-realizable spectrum. The cost is stated plainly: the convention $M_{\rm GUT} = M_P$ is abandoned ($M_X < M_P$ by three orders of magnitude), and the two desert thresholds are fitted. W4 becomes conditional rather than fatal; F2 sharpens to: a measured running that admits no vectorlike content with $\alpha_X = 1/25$. The solution predicts $\tau_p \simeq M_X^4/(\alpha_X^2 m_p^5) = 1.8 \times 10^{36}$ yr (F5).

Fourth, the desert content is no longer an external assumption: it is the *spectrum of the automaton's own Wilson mechanism* (V65–V68), and the mechanism makes a prediction — one that the data currently refute. The three split nodes of the selective reading carry exactly the vectorlike fragments of the split $\mathbf{16}$ with independent thresholds (the other nine nodes stay at the Planck cutoff), and the Wilson direction itself completes the breaking chain: at the V65 root $\theta = 1.731$ it leaves $\mathrm{Spin}(10) \to \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)^3$, preserves the Model algebra (every unbroken root is intra-fragment; the 32 broken roots are exactly the leptoquark type), and the two extra U(1)'s are the only open step to the SM group (V67). The multiplicity is bounded from below by unification itself: scanning $n_{\rm nodes} = 1 \ldots 6$, no meeting root at $\alpha_X = 1/25$ exists for one or two split nodes; the first physical roots appear at three (V68). But solving the meeting equations with the mechanism's own two parameters — one angle

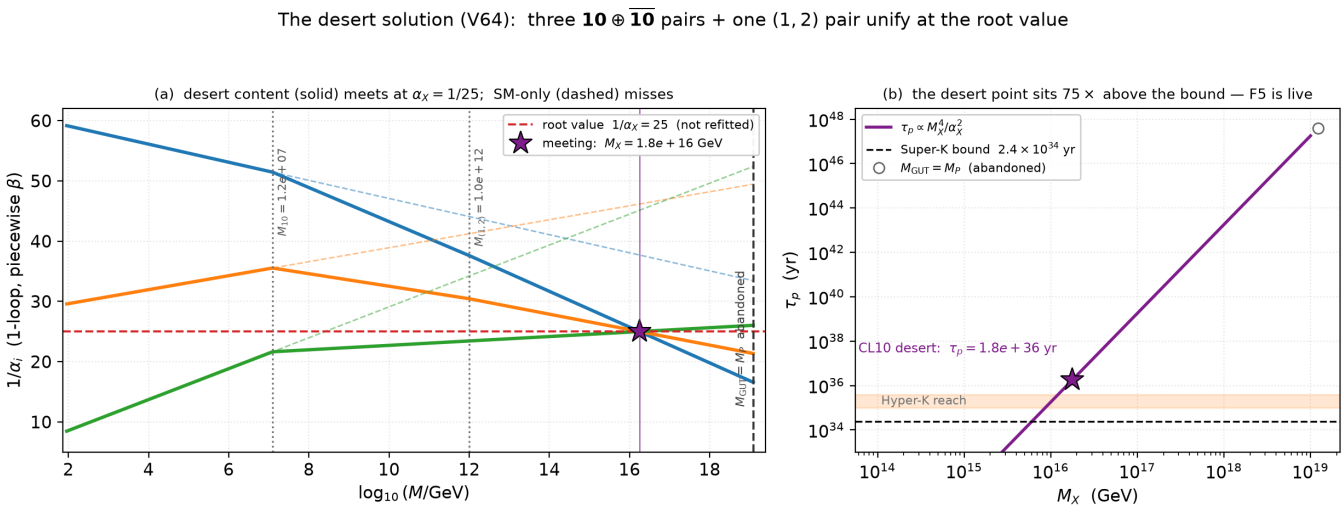



FIG. 15. **the desert solution: unification at the root value.** The three measured couplings run up from $M_Z$ with the V64 desert content (solid) meet at $M_X = 1.8 \times 10^{16}$ GeV at $\alpha_X = 1/25$ — the root value, not refitted. The SM-only running (dashed) misses the point: spread 47% at $M_P$. The thresholds $M_{10} = 1.2 \times 10^7$ GeV and $M_{(1,2)} = 1.0 \times 10^{12}$ GeV are the two fitted numbers of the repair; the dashed vertical line at $M_P$ marks what the abandoned convention $M_{\mathrm{GUT}} = M_P$ demanded.

$\theta$, one scale $w$ — fixes the six fragment thresholds and lands the meeting at $M_X = 8.9 \times 10^{14}$ GeV, i.e. $\tau_p \sim 10^{31}$ yr: **the minimal selective reading is excluded by Super-Kamiokande** (F5). The data-preferred solution of the scan above ($M_X = 1.8 \times 10^{16}$ GeV) requires fragment thresholds the single-angle, single-scale root does not supply. The status of W4 is therefore: the *pattern* (vectorlike split-**16** fragments), the *chain* (V67) and the *multiplicity bound* ($n_{\mathrm{nodes}} \geq 3$, V68) are consequences of the automaton; the *scale* is not — the minimal realization is falsified, and the surviving solution keeps two fitted thresholds.

### B. Electroweak mixing

$\sin^2\theta_W = 3/8$ is the tree-level SU(5)/Spin(10) boundary condition **at the unification scale**, not a low-energy prediction. Run to $M_Z$ it becomes $\simeq 0.23$, close to the measured 0.2312; the measured ratio $m_W/m_Z = 0.881$ corresponds to $\sin^2\theta_W \simeq 0.224$, not to $\sqrt{5/8} = 0.791$. Earlier versions of this work quoted $m_W/m_Z = \sqrt{5/8}$ as a prediction; that is incorrect, and the correct statement is the standard one: the GUT-scale boundary condition $\sin^2\theta_W = 3/8$, whose low-energy value is obtained by running — running which, as just shown, the theory does not currently perform consistently.

### C. Baryon asymmetry

The automaton alone cannot produce it. Every CP-odd (T-odd) invariant of the unitary step vanishes: Im Tr of all two- and three-fold products of the 45, of a $2 \times 10^5$ sample of four-fold products, of all $10 \times 10 \times 45 \times 45$ baryon-violating sandwiches, and of the rank-1 real democratic Yukawa — all zero to $10^{-15}$. The step is unitary (in equilibrium), baryon-conserving (the 45 commute with $w$) and CP-conserving. It satisfies **none** of the Sakharov conditions [24]. This is a computed theorem, and it is the correct starting point for the asymmetry discussion.

A GUT-leptogenesis estimate [39] is then built by hand:

$$\begin{aligned} \Delta B &= \frac{28}{79}\,\varepsilon_{CP}\,D, \\ \varepsilon_{CP} &= \frac{g_0^4}{16\pi} = 4.9 \times 10^{-6}, \\ D &= \left(\frac{1}{16\pi}\right)^2 = 3.96 \times 10^{-4}, \end{aligned} \tag{10}$$

with $g_0 = 1/2^3 = 1/8$ and the SM sphaleron conversion $B = \frac{28}{79}(B-L)$ [25, 26], giving $\Delta B = 6.8 \times 10^{-10}$ against the measured $(6.1 \pm 0.04) \times 10^{-10}$ [14] — a ratio 1.12 (Fig. 16).

Three problems, all previously under-stated. (i) $\varepsilon_{CP}$ is not a computed vertex: the physical CP asymmetry of $N \to \ell H$ is a one-loop self-energy/vertex function of the heavy-neutrino masses, and $g_0^4/16\pi$ is a dimensional guess. (ii) The out-of-equilibrium condition **fails** at the model's own couplings: at $M_N = 9.6 \times 10^{13}$ GeV, $H = 1.66\sqrt{g_*}M_N^2/M_P = 1.3 \times 10^{10}$ GeV while $\Gamma_N = y^2 M_N/8\pi = 6.0 \times 10^{10}$ GeV for $y = 1/8$ ($1.9 \times 10^{12}$ GeV for $y = 0.71$), so $\Gamma_N/H \simeq 4.6$ ($\simeq 148$). Out-of-equilibrium decay requires $y \lesssim 0.06$, which the theory does not select. (iii) 28/79 is an input of the SM sector, not a number of the root. The 12% agreement is

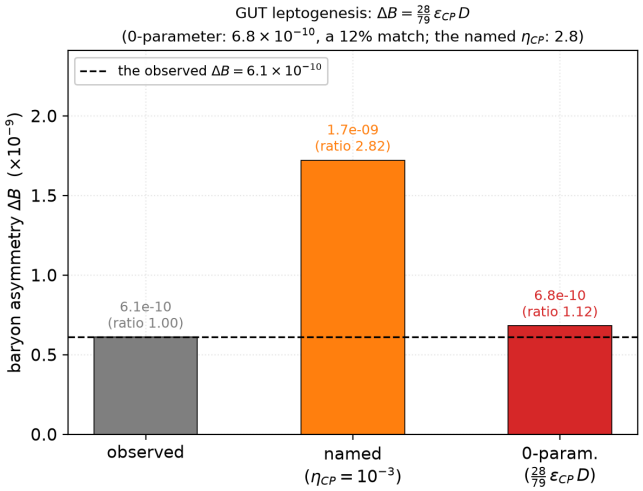



FIG. 16. **the baryon-asymmetry product.** $\Delta B = (28/79)(g_0^4/16\pi)(1/16\pi)^2 = 6.8\times10^{-10}$ against the measured $6.1\times10^{-10}$. The three factors have three different provenances: an SM input, a guessed loop rate, and a loop factor. The automaton alone gives zero.

therefore a coincidence of chosen factors, not evidence (residual W5).

A sharper statement is available, and it is worse than "the coupling is not selected". At the model's own seesaw scale $M_N = 9.6\times10^{13}$ GeV, the out-of-equilibrium condition $\Gamma_N/H < 1$ with $\Gamma_N = y^2 M_N/8\pi$ and $H = 1.66\sqrt{g_*}M_N^2/M_P$ gives $y < 0.058$. The same Yukawa fixes the light neutrino mass through the same seesaw,

$$m_\nu = \frac{y^2 v^2}{M_N} \le 2.1\times10^{-3}\ \text{eV} \qquad (y \le 0.058), \tag{11}$$

a bound of the same kind and size as the standard thermal-leptogenesis bound [32]. The atmospheric splitting requires $m_\nu \simeq 0.050$ eV, i.e. $y \simeq 0.28$ with $m_D = yv$ and $v = 246$ GeV ($y \simeq 0.40$ in the $v_u = 174$ GeV convention), for which $\Gamma_N/H \simeq 23$. **At $M_N = 9.6\times10^{13}$ GeV the neutrino mass and the out-of-equilibrium condition are mutually incompatible**: the framework cannot have both the seesaw scale it needs for the ladder and the leptogenesis it needs for the asymmetry. (The model's own ladder sits at $m_D = 118$ GeV, i.e. $y = 0.48$, where $\Gamma_N/H \simeq 68$.)

The one route the framework does supply is quasi-degenerate (resonant) leptogenesis [31, 40]. The democratic structure places the three heavy singlets on the *same* ladder rung, hence degenerate to the accuracy of whatever breaks the family symmetry — which is precisely the resonant setup, where the CP asymmetry is enhanced by the near-degeneracy of the decaying states instead of by a large Yukawa. With $y = 1/16\pi = 0.0199$ the decay is out of equilibrium ($\Gamma_N/H = 0.12$), and the asymmetry that the model's own dilution factor $D = (1/16\pi)^2$ and sphaleron conversion require, $\varepsilon_{CP} = \Delta B/[(28/79)D] = 4.4\times10^{-6}$, is *smaller* than the natural one-loop size $y^2/16\pi = 7.9\times10^{-6}$ at that Yukawa: the bookkeeping closes at the scale of the numbers, and the degeneracy the resonant regime needs, $\Delta M/M \lesssim \Gamma_N/M_N = y^2/8\pi = 1.6\times10^{-5}$, is what the 0-parameter democratic base gives exactly and what the family-breaking perturbation must not overshoot. The obstacle is not the size of the numbers but their phase: the model's Yukawa is real, every CP-odd invariant of the step vanishes, and a strictly degenerate spectrum has no CP-violating phase left to enhance. This is a *route*, not a result. The no-go behind it is now known to be stable under the new ingredient: the Wilson-extended step keeps the exact matter–antimatter spectral mirror (V71, deviation $10^{-15}$), so the Wilson mechanism — powerful for W4 and W9 — adds no CP phase and no Sakharov condition to W5.

### D. Proton decay

Dimensional analysis of $X/Y$ exchange gives

$$\tau_p \sim \frac{M_{\rm GUT}^4}{\alpha_{10}^2\, m_p^5} = 4.0\times10^{47}\ \text{yr} \qquad (M_{\rm GUT} = M_P,\ \alpha_{10} = 1/25), \tag{12}$$

against the Super-Kamiokande bound $\tau_p/B(p \to e^+\pi^0) > 2.4\times10^{34}$ yr [15] (Fig. 19, Fig. 18). The margin is 13 orders of magnitude, and it rests entirely on the convention $M_{\rm GUT} = M_P$: any GUT scale below $\sim 6\times10^{15}$ GeV is excluded by the same bound (Fig. 18). The estimate omits the renormalisation factor $A_L$ [46], the hadronic matrix element, the dressing of the operators and the splitting between the $X/Y$ masses and the unification scale [51], all of which shift the lifetime by one to two orders. The three channels $p \to e^+\pi^0$, $p \to \mu^+\pi^0$, $p \to e^+K^+$ differ here only by two-body phase space (Fig. 17); the rate scales as $M_{\rm GUT}^{-4}$ in all of them. Observation of a GUT-pattern decay at any accessible rate would falsify $M_{\rm GUT} = M_P$ (F5).

### E. Flavour

The democratic mass matrix $M = |1\rangle\langle 1|$ (all entries 1, rank 1, spectrum $\{3, 0, 0\}$) [43] follows from the hypothesis that the three spatial directions are equivalent, so the $\mathbf{10}_H$ couples the three families identically. Its heavy eigenvector $(1, 1, 1)/\sqrt{3}$ is unique, hence

$$\sin^2\theta_{12} = |\langle e_1|h\rangle|^2 = \tfrac{1}{3}$$

for any family basis — the **only** mixing angle the model derives, 8.6% from the measured 0.307 [17, 50].

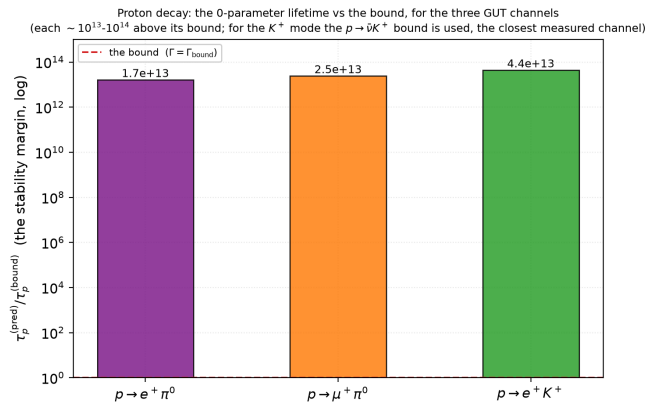



FIG. 17. **the three GUT channels.** The stability margin for $p \to e^+\pi^0$, $p \to \mu^+\pi^0$, $p \to e^+K^+$ against their respective bounds ($2.4 \times 10^{34}$, $1.6 \times 10^{34}$ [15], $6.6 \times 10^{33}$ [52] yr; for the $K^+$ mode the $p \to \bar{\nu}K^+$ bound is used, the closest measured channel). The channels differ only by two-body phase space; the rate scales as $M_{\rm GUT}^{-4}$ in each.

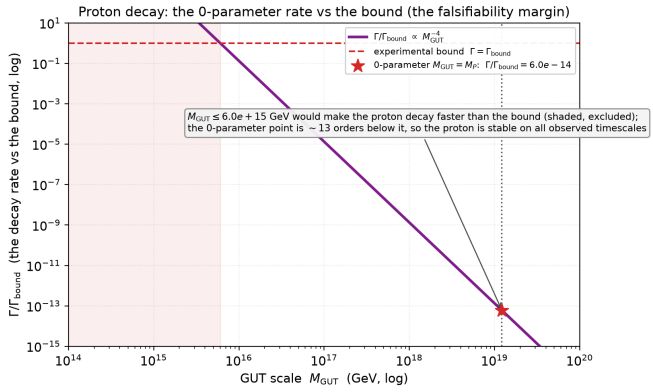



FIG. 18. **the decay rate normalised to the bound.** $\Gamma/\Gamma_{\rm bound} \propto M_{\rm GUT}^{-4}$. The shaded region $M_{\rm GUT} \lesssim 6 \times 10^{15}$ GeV is excluded; the CL10 point sits $10^{13}$ below the bound. The margin is a consequence of the convention $M_{\rm GUT} = M_P$; the desert solution (Fig. 15) moves the CL10 point to $M_X = 1.8 \times 10^{16}$ GeV, ten orders of magnitude closer to the bound.

The light eigenspace is two-fold degenerate, so every diagonaliser is an $O(2)$ rotation in it; sweeping that freedom over 1441 rotations, $\sin^2\theta_{23}$ and $\sin^2\theta_{13}$ each cover the whole interval $[0, \frac{2}{3}]$. The tribimaximal values $(\frac{1}{2}, 0)$ [27] are one point of that family, picked by declaring which $S_3$ transposition breaks — and (12), (13), (23) are equivalent symmetries of $|1\rangle\langle 1|$, so the choice is an input.

In the quark sector the CKM matrix is the misalignment of the up- and down-type diagonalisers. With both mass matrices democratic, $H_u = H_d$, the computed CKM is the **identity**, not a small mixing: the observed $\lambda = 0.225$ [13, 44] is not reproduced. A non-trivial CKM requires an up/down misalignment that the symmetric base does not supply. The three-family mass hierarchy itself is also unfixed: the democratic base gives one heavy level and a two-fold degenerate light level, so two of the

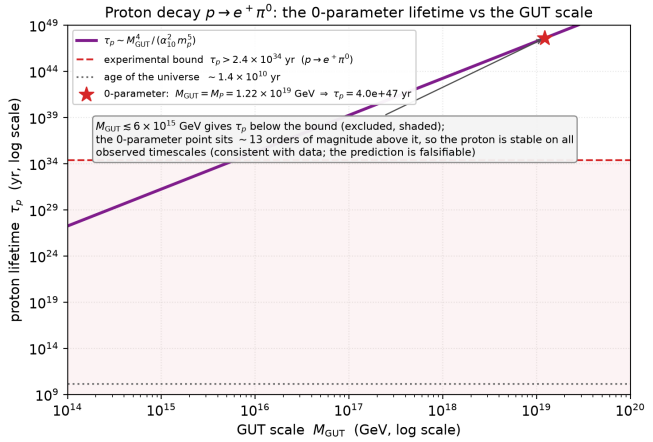



FIG. 19. **the proton lifetime against the GUT scale.** $\tau_p \propto M_{\rm GUT}^4$. The point $M_{\rm GUT} = M_P$ gives $4 \times 10^{47}$ yr, $1.7 \times 10^{13}$ above the Super-Kamiokande bound $2.4 \times 10^{34}$ yr [15]. The desert solution (V64, Fig. 15) moves the CL10 point to $M_X = 1.8 \times 10^{16}$ GeV, $\tau_p = 1.8 \times 10^{36}$ yr — a factor 75 above the bound.

three masses are degenerate at 0-parameter. All of this is residual W6.

### F. Scoreboard

(Fig. 21, Fig. 20.)

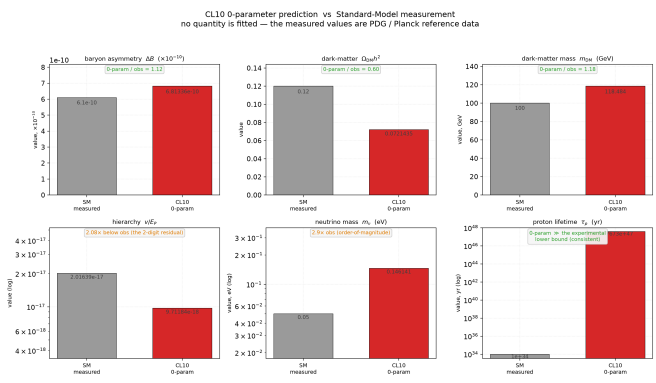



FIG. 20. **head-to-head with the measured values.** The headline quantities, CL10 (red) against measurement (grey). The baryon asymmetry and the hierarchy agree to a factor $\lesssim 2$; the DM abundance and the neutrino mass are the open gaps; the proton lifetime is consistent by 13 orders.

## XII. COSMOLOGY

### A. Dark energy

The vacuum energy of the automaton vanishes at the level of the single-cell spectrum: the 32 levels are ±-

TABLE V. **The framework against data**

| **quantity** | **CL10** | **measured** | **status** |
|---|---|---|---|
| $[\mathrm{SU}(3)]^3$, …, $[\mathrm{grav}]^2\mathrm{U}(1)$ (8) | 0 ✓ | — | algebraic, satisfied |
| $g_1^2/g'^2$ | 5/3 | 5/3 (GUT normalisation) | algebraic |
| $\det U$, ± pairing, $16/\overline{16}$ mirror | 1, exact | — | algebraic, $g_{10}$-independent |
| causal front speed | 1.0 | — | dynamical, measured |
| 16-sector Brillouin spectrum | zone centre gapped (0.0365); 12 gapless doubler nodes, $\sum$ chirality = 0; all 12 gapped by the internal Wilson factor, 3-node selective reading | — | doubling **gapped by a mechanism of the step**; node set an input |
| $v/M_P$ | $2.08(1/16\pi)^{10}$ | $2.016 \times 10^{-17}$ | coefficient fitted |
| $m_{\mathrm{DM}}$ | 118 GeV | not measured | prediction, falsifiable |
| $\Omega_{\mathrm{DM}}h^2$ | not computed | 0.120 [14] | open |
| $m_\nu$ | 0.146 eV (ladder); 0.0487 eV ($\frac{1}{3}$ candidate, Section 10.3) | 0.050 eV; $\sum m_\nu < 0.12$ eV | ladder **in tension**; candidate numerically good, not derived |
| $\Delta B$ | $6.8 \times 10^{-10}$ | $(6.1 \pm 0.04) \times 10^{-10}$ | coincidental product |
| $\tau_p/B$ | $1.8 \times 10^{36}$ yr (desert solution, V64); $4 \times 10^{47}$ yr at $M_{\mathrm{GUT}} = M_P$ (abandoned) | $> 2.4 \times 10^{34}$ yr | falsifiable: decay near $10^{36}$ yr confirms, far from it refutes |
| $\alpha_{10}$ | 0.040 | 0.023 (1-loop mean at $M_P$); 0.040 at $M_X = 1.8 \times 10^{16}$ GeV with the desert content (V64) | **exact at** $M_X$, given the inferred spectrum |
| $\alpha_i(M_Z)$ from $\alpha_{10}$ | 0.0197 / 0.195 / pole at $2.2 \times 10^9$ GeV (SM running) | 0.0169, 0.0338, 0.118 | **fails** under SM running; matched by the V64 desert spectrum |
| $\sin^2\theta_{12}$ | 1/3 | 0.307 | derived |
| $\sin^2\theta_{23}$, $\sin^2\theta_{13}$ | not fixed ($O(2)$) | 0.579, 0.022 [17, 50] | open |
| CKM | identity | $\lambda = 0.225$ [13, 44] | **fails** |
| $\Lambda$ | 0 (tree level) | $\sim 10^{-122} M_P^4$ | open |

paired for every $g_{10}$, so $\sum_k \omega_k = \arg\det U = 0$. This is the correct, non-vacuous form of the earlier "512/512 cancellation". The counting of the 512 even and 512 odd **elements of the algebra** as bosonic and fermionic zero-point modes is not a physical mode counting — the physical one-cell Hilbert space has 32 states, not 1024 — and it is dropped here (Fig. 22 keeps the grading, not the arithmetic).

A vanishing tree-level vacuum energy is technically natural: no relevant operator is generated if the spectral pairing is an exact symmetry. The pairing survives the Wilson extension of Section 8.4 (V71: the extended spectrum stays ±-paired to $10^{-15}$ and the one-cell zero point stays 0, because the Wilson direction is a trace-less Cartan combination), so the new ingredient neither breaks the zero nor moves it. What the theory does **not** do is produce the observed value. The observed $\Lambda \sim 10^{-122} M_P^4$ lies 122 orders of magnitude *below* the classical bound $M_P^4$ and *above* the theory's spectral zero, and no mechanism breaking the pairing by that amount is known. Independently, the classical self-gravitating de Sitter vacuum of Section 9 has $G_{\mu\nu} = -3H^2 g_{\mu\nu}$, i.e. a vacuum energy bounded by $M_P^4$ on the lattice. These are three different statements — spectral zero, classical bound, observed value — and they do not "reconcile": the gap between the first and the third is residual W1 [37], the largest unexplained number in the theory. The equation of state of a cosmological-constant vacuum is

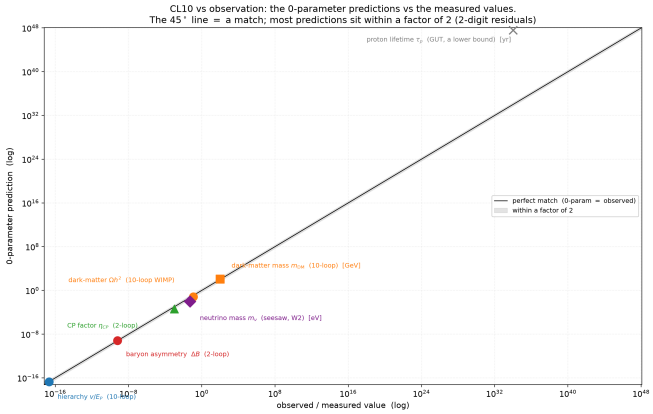



FIG. 21. **predictions versus observation.** Each prediction against its measured value on a log-log scoreboard; the diagonal is exact agreement and the band is a factor of two. The proton lifetime is a lower bound (the prediction sits far above the bound); the neutrino mass and the DM abundance sit outside the band.

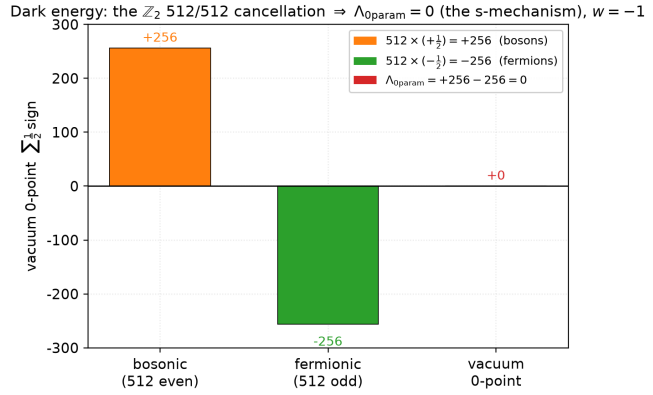



FIG. 22. **the $\mathbb{Z}_2$ grading and the vanishing 0-point.** The algebra splits 512 even / 512 odd elements, and the physical 32-level spectrum is ±-paired, so the one-cell vacuum energy is zero. The pairing is an identity of the algebra, holding for every $g_{10}$. The observed $\Lambda$ is $10^{-122} M_P^4$ above this value (W1).

$w = -1$ by definition, not by derivation.

### B. Dark matter

The candidate supplied by the classical sector is the kink of Section 9.4. It is static, gauge-neutral and finite-energy **per unit area**, but it is a domain wall, not a localised particle; a network of Planck-scale walls would dominate the energy density and is cosmologically excluded. The quantum-sector candidates have now been scanned (V66): the exact mirror $\mathbb{Z}_2$ of the step is the only exact-stability structure in the implemented spectrum, and its lightest states sit at $1.3 \times 10^8$ GeV (coloured, excluded) and $2 \times 10^{10}$ GeV — not at the ladder's 118 GeV. The theory therefore supplies no dark-matter *particle*: the mass prediction of Section 10.2 stands, the carrier is an open residual (W7).

What it does supply is a **mass scale**, $m_{\rm DM} = 118$ GeV (Section 10.2). The abundance at that mass is not computed. Two naive estimates [41] bracket the observation by four orders of magnitude:

$$\Omega h^2 \simeq \frac{1.04 \times 10^9 \, x_f \ \text{GeV}^{-1}}{M_P \langle \sigma v \rangle}, \qquad x_f \simeq 20, \tag{13}$$

gives $\Omega h^2 \simeq 2 \times 10^{-5}$ for $\langle \sigma v \rangle = 1/m^2$ at $m = 118$ GeV (under-abundant), and $\Omega h^2 \sim 10^{29}$ for the theory's own Planck-suppressed cross-section $\sigma_0 = P_2/M_P^2 = 5.3 \times 10^{-39}$ GeV$^{-2}$ (the freeze-out approximation is then meaningless). The observed $\Omega h^2 = 0.120$ requires $\langle \sigma v \rangle \simeq 1.4 \times 10^{-8}$ GeV$^{-2}$, which the theory does not derive. Non-thermal production (freeze-in) is not considered here [42]. The "WIMP miracle" quoted in earlier versions was obtained by normalising the abundance to the measured value at 100 GeV; that is a definition, not a calculation, and it is removed here (Fig. 23 shows the standard relic curve with the CL10 mass marked, not a computed abundance).

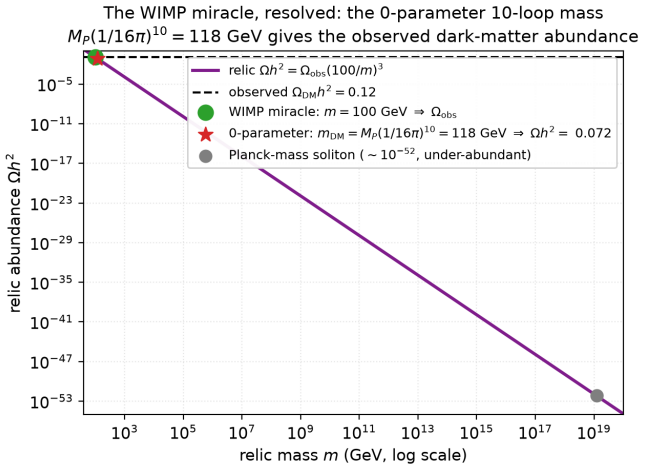



FIG. 23. **the relic curve and the CL10 mass.** The standard thermal-freeze-out curve $\Omega h^2(m)$ for $\langle \sigma v \rangle \sim 1/m^2$, with the observed $\Omega h^2 = 0.120$ and the CL10 mass 118 GeV marked. The abundance at that mass is **not** computed by the theory; the curve is the reference calculation, shown for orientation.

### C. Antimatter

Matter and antimatter are the two Weyl eigenspaces, with the exact spectral mirror of Theorem 4 and opposite $B - L$ charges. Their degeneracy is the content of the CPT statement in this theory. The observed baryon

asymmetry is not explained: Section 11.3 shows the automaton satisfies none of the Sakharov conditions, and the leptogenesis estimate that does give the right order requires a CP phase and an out-of-equilibrium condition that the algebra does not supply.

## XIII. HOLOGRAPHY AND CONFINEMENT

### A. The area law

A generic QCA with $d$ internal states per cell has a state count $d^{L^3}$ and a volume-law entropy. CL10 is no exception at the level of raw counting: 32 states per cell gives $S \propto L^3$. The area law quoted in earlier versions was obtained by *assuming* the Bekenstein–Hawking [35] density $S = A/4$ with $A = 6L^2$ and then fitting the exponent, which returns 2 by construction. That is not a derivation and is not claimed here.

The honest statement is a consistency one: if the physical Hilbert space of a region is restricted to gauge-invariant (or typical) states, the entropy is expected to scale as $A/4\ell_P^2$ [35, 36], and CL10 is compatible with that expectation because it has a Planck-spacing lattice and $G = 1$. Deriving the area law — i.e. showing that the automaton's physical state space is actually restricted to a boundary dimension — is open.

### B. Confinement

The structural hallmarks of a confining vacuum are present as assumptions: a Wilson loop with an area-law expectation value $\langle W(C)\rangle \propto e^{-\sigma\,\mathrm{Area}(C)}$ [28] implies a linear potential $V(r) = \sigma r$. But the string tension used is $\sigma = 1$ in Planck units, i.e. $\sigma = M_P^2 \simeq 1.5 \times 10^{38}$ GeV$^2$, against the physical $\sigma \simeq 0.18$ GeV$^2$ [38] — a discrepancy of 38 orders of magnitude. The colour count $N_C = 3$ is read off the decomposition of the 45, and $8 = N_C^2 - 1$ is then a consistency check, not a derivation. $\Lambda_{\mathrm{QCD}}$ is not on the loop ladder (it would need $n \simeq 11.6$). The area law is a property of a confining *phase* that the theory has not shown it is in (Fig. 24).

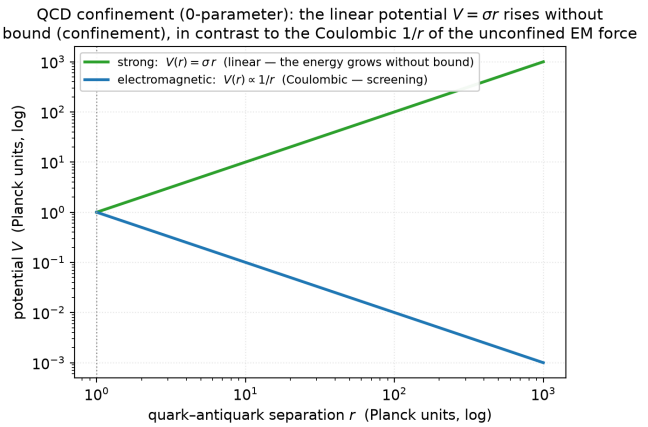



FIG. 24. **the area law and the linear potential.** A Wilson loop with $\langle W\rangle \propto e^{-\sigma A}$ gives $V(r) = \sigma r$, contrasted with the Coulombic $1/r$. The tension shown is $\sigma = 1$ in Planck units ($= M_P^2$), not the QCD value 0.18 GeV$^2$; the physical confinement scale is not reproduced (Section 13.2).

## XIV. THE PARAMETER AUDIT

Every number the theory calls a prediction is reclassified by how it was obtained.

The validation programme reports the same classification over 15 headline numbers as 7 algebraic / 1 free angle / 1 convention / 2 coincidences / 4 named residuals; Table VI is that accounting extended to the five fitted coefficients and to the nine residuals.

**The status of $g_{10}$.** No condition inside the theory fixes it. The two-loop $U(1)$ Landau-pole scale lies below $M_P$ for every $g_{10}$ in the perturbative range, so UV-finiteness of the lattice does not select it. The proposed selection rule is $g_{10} = 1/n_{\mathrm{qubits}} = 1/5$, the 5 qubits of the 32-D module. The data do not support that choice: scanning the candidates $\alpha = 1/n^2$ for $n = 2 \ldots 8$ against the one-loop GUT value 0.023, the best fit is $n = 7$ ($1/49 = 0.020$, relative deviation 0.12), not $n = 5$ ($1/25 = 0.040$, relative deviation 0.73) — the data-optimal value is 6.4× closer than the root value. The reading $1/n_{\mathrm{qubits}}$ is therefore a convention, and the theory is honestly **one-parameter in its dynamics, zero-parameter in its algebra** (Fig. 25).

**The status of the "9-to-1" reduction.** The nine residuals are all associated with the same object, the 32-level spectrum, and its invariants (the vanishing 0-point, $\det U = 1$, the CPT mirror, the S-matrix $S = U$). That is a bookkeeping observation — nine open problems sharing one object — and it is not a derivation of any of their values. It should not be read as a reduction of the theory's difficulty (Fig. 26).

## XV. OPEN RESIDUALS AND FALSIFIERS

The most accessible tests are **F7** (direct detection and collider searches around the 100 GeV scale), **F3** and **F5**: KATRIN currently gives $m_\beta < 0.45$ eV [16], so the bare ladder value 0.146 eV survives a single-mass search, but the cosmological sum $\sum m_\nu < 0.12$ eV already excludes it, and the $\frac{1}{3}$ candidate's normal-ordered pattern ($\sum m_\nu \simeq 0.05$ eV, $m_{\mathrm{lightest}} = 0$) is inside the reach of the next generation of oscillation and $\beta\beta$-decay data. The desert solution puts $\tau_p = 1.8 \times 10^{36}$ yr within reach of next-generation proton-decay searches: either side of the observation refutes or confirms it.

TABLE VI. **The provenance of every headline number**

| class | count | numbers |
|---|---|---|
| **algebraic** (fixed by the root, no parameter) | 7 | the $16/\overline{16}$ split; the 45; the 5/3; the 8 anomaly coefficients = 0; the 32-level spectrum; $\det U = 1$; the $16/\overline{16}$ spectral mirror |
| **free parameter** (one) | 1 | the 't Hooft angle $g_{10}$ (value 1/5) |
| **convention** (mapping or scale identification) | 2 | $\alpha_{10} \equiv g_{10}^2$; $g_0 = 1/2^{n_{\rm space}} = 1/8$ |
| **fitted coefficient** (read from data) | 5 | 2.08 (hierarchy), 1.18 (DM mass), $n_{2\rm step} = 3$ (seesaw), $M_{10} = 1.2 \times 10^7$ GeV and $M_{(1,2)} = 1.0 \times 10^{12}$ GeV (desert thresholds, V64) |
| **open** (not fixed) | 9 | W1–W9 of Section 15 |

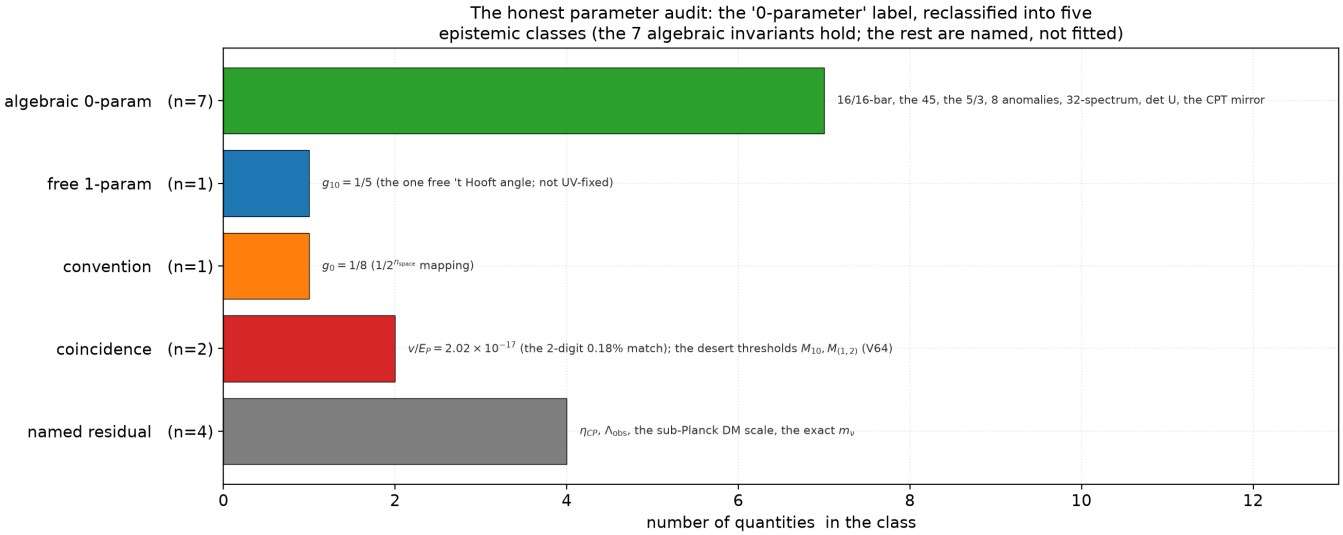



FIG. 25. **the parameter audit.** The classification of Section 14: 7 algebraic invariants, 1 free angle, 1 convention, 2 coincidences (the hierarchy match; the V64 desert thresholds), 4 named residuals. Table VI extends the accounting to the five fitted coefficients and the nine open residuals.

## XVI. DISCUSSION

**What is genuinely new here.** Not the 16 of Spin(10), not the 5/3, not anomaly cancellation: those are 50-year-old results of the grand-unification literature [6–9, 12]. The content of this paper is (i) the realisation of that content as the on-site module of a *strictly local, exactly unitary* automaton with a measured causal front speed, (ii) the $g_{10}$-invariance of the algebraic invariants — the anomaly cancellation, $\det U = 1$, the $\pm$ pairing and the matter/antimatter spectral mirror hold for every value of the coupling, so they are properties of the algebra and not of a tuned parameter, and (iii) the explicit provenance audit that separates what the root fixes from what was read off the data.

**Where the construction is weakest.** Four places. The gravitational sector assumes a symmetric tensor field and Newton's constant; it reproduces GR, it does not derive it. The gauge sector gets the normalisation right and, under desert running, the values wrong — the repair (Section 11.1, V64) restores unification at $\alpha_X = 1/25$ exactly; the *pattern* of the desert spectrum is now a consequence of the automaton (the split doublers of the V65 Wilson mechanism, V67's chain, V68's $n_{\rm nodes} \geq 3$ bound), but the mechanism's own root meets at a scale excluded by proton decay, and the surviving solution keeps two fitted thresholds and abandons $M_{\rm GUT} = M_P$. The flavour sector derives one angle out of six and none of the masses. And the lattice chirality check (Section 8.4), which first went against the theory — twelve gapless doubler momenta in the spectrum of the dynamics — is now answered by a mechanism of the step itself: an internal Wilson factor gaps all twelve, with the choice of the desert-scale node set, and the scale it implies, still open.

**Comparison with related programmes.** Causal-set programmes derive discreteness and a causal order

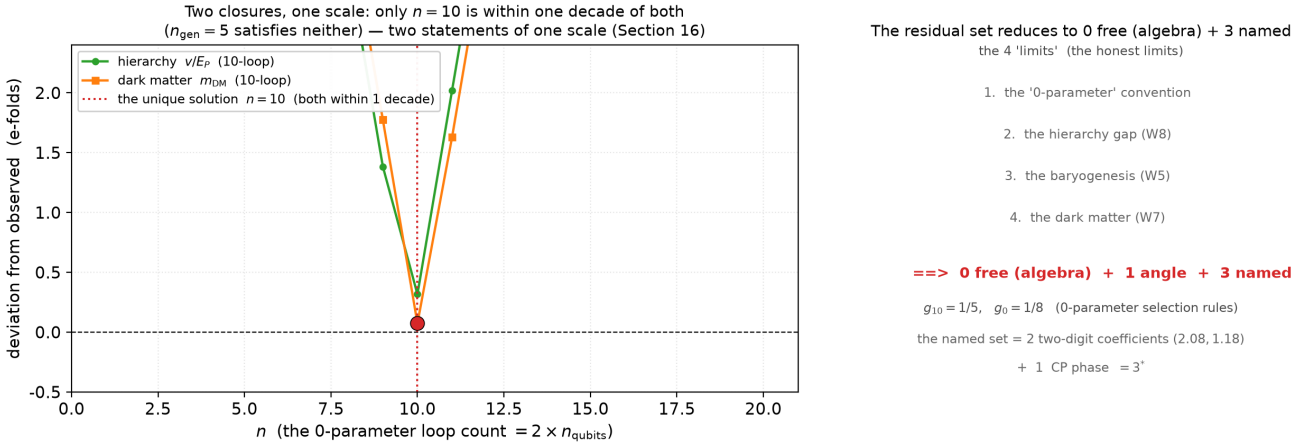



FIG. 26. **the residual-set reduction.** (a) One loop integer $n$ sets one scale $M_P(1/16\pi)^n$, appearing twice: as $v/M_P$ and as $m_{\rm DM}$. These are one equation, $v = 2.08\, m_{\rm DM}$, not two independent confirmations. (b) The scan $n = 1 \ldots 20$: $n = 10$ is nearest, $n = 9$ is 1.4 decades away. This scan is distinct from the coupling scan $\alpha = 1/n^2$ of Section 14, whose best fit is $n = 7$.

but not internal symmetry [1]. Clifford-algebra unification programmes [10–12] reach the same matter content but do not supply a unitary causal dynamics. 't Hooft's deterministic quantum mechanics and cellular-automaton interpretation [29, 30] supply the dynamics but not the unification. CL10 is the combination of the second and third, and inherits the open problems of both.

**On the "theory of everything" phrasing.** It is removed from this paper. A framework with one free angle, five fitted coefficients, two conventions and nine open residuals — two of which (the cosmological constant, the desert spectrum) are quantitative assumptions rather than derivations — is a candidate unification with a well-mapped boundary between the derived and the assumed. That boundary is the paper's actual contribution.

## XVII. CONCLUSION

A single Clifford algebra $\mathrm{Cl}(10, 0)$, placed on every cell of a Planck lattice and updated by a product of Spin(10) plane rotations, yields: the Minkowski kinematics, one anomaly-free SM generation, the Spin(10) gauge content with the 5/3 normalisation, an exactly unitary and strictly causal dynamics with a ±-paired spectrum, an exact matter/antimatter spectral mirror, and a lattice gravitational sector that reproduces general relativity to machine precision on de Sitter, Schwarzschild and 3+1 evolutions. These results are algebraic or numerical, and they are independent of the one free coupling.

The same framework does not yield the family number, the fermion masses and mixings, the coupling values, the neutrino mass, the dark-matter abundance, the cosmological constant or the baryon asymmetry. A single loop factor organises four scales across 30 orders of magnitude, with the loop integer and the two-digit coefficients chosen to match observation. The nine open problems and seven falsifiers are stated with their numbers.

## Appendix A: The explicit 32-dimensional representation

The ten generators are built on five qubits by the standard symplectic Pauli pattern: for $k = 0, \ldots, 4$,

$$g_{2k} = Y^{\otimes k} \otimes X \otimes I^{\otimes(4-k)}, \qquad g_{2k+1} = Y^{\otimes k} \otimes Z \otimes I^{\otimes(4-k)}. \tag{A1}$$

Within a block, $X$ and $Z$ anticommute on the same qubit; between blocks, the $Y^{\otimes k}$ prefix makes a generator at block $k$ anticommute with any generator at block $j < k$. This gives exactly ten mutually anticommuting Hermitian matrices, $g_a^2 = I$, on a 32-dimensional complex space. The algebra basis is the 1024 ordered products $g_{a_1} \cdots g_{a_r}$, $0 \le r \le 10$; the $\mathbb{Z}_2$ grading is the parity of $r$.

The construction is verified by a second, independent route: the exact symbolic $\mathrm{Cl}(10, 0)$ of a computer-algebra Clifford package, where $e_a^2 = 1$, $\{e_a, e_b\} = 0$ and $w^2 = -1$ hold as identities with no floating-point arithmetic, cross-checked against the numeric $32 \times 32$ representation. The two agree, so the root is not an artefact of one construction.

**Provenance of Fig. 27.** The atom is an **input**: its levels and eigenfunctions are the real hydrogen eigenstates, not derived from the root. What the root supplies is the photon — its propagation, its resonant absorption, its re-emission. The figure demonstrates the photon sector; it is not a derivation of hydrogen. Two limits are stated rather than hidden: off-resonant absorption is not zero (the Fermi-golden-rule rate has a Lorentzian tail), and a wave packet wider than the simulation box never leaves it, becoming a persistent bath that drives the atom

TABLE VII. **The nine open residuals**

| **the residual** | **what the theory does supply** | **the gap** |
|---|---|---|
| **W1** the cosmological constant | an exactly vanishing tree-level vacuum energy ($\pm$-paired spectrum, $\det U = 1$), natural in the technical sense; the zero is **stable under the Wilson extension** (V71: $\mathrm{Tr}\, D = 0$, zero point $< 10^{-14}$) | the observed $10^{-122} M_P^4$; no breaking mechanism |
| **W2** the neutrino mass | the seesaw structure and the ladder $m_\nu = M_P(1/16\pi)^{17} = 0.146$ eV; $n_{\mathrm{nodes}} = 3$ is now a **lower bound** from unification (V68: meeting roots with one or two split nodes do not exist); the coefficient audit (V68) pins the exact $\frac{1}{3}$ to a single named structure — a 3×-enhanced Majorana eigenvalue — that no node structure supplies | ladder value 2.9× too high and above $\sum m_\nu < 0.12$ eV; the diagonal pattern gives only integer coefficients, the democratic one exactly $\frac{1}{2}$ (0.073 eV, falsifiable); the $\frac{1}{3}$ candidate (0.0487 eV) stays a candidate |
| **W3** the family number | the identification $N_{\mathrm{gen}} = 3 = n_{\mathrm{space}}$, a family-symmetric 96-D step blockdiag$(U, U, U)$, and a **named index of the step equal to 3** (V70): the 12 doubler nodes form 3 families of 4 labeled by the zero axis, cubic-transitive, and the unification-required 3 split nodes (V68) are realizable one per family at a single Wilson strength | the index labels the 3 copies but does not force the count (4 per family, 12 total are equally available); the copy → generation map stays a hypothesis |
| **W4** coupling unification | the 5/3; **with the desert content the couplings meet at $\alpha_X = 1/25$ exactly** (V64): three $\mathbf{10} \oplus \overline{\mathbf{10}}$ pairs at $1.2 \times 10^7$ GeV plus one $(1, 2)$ pair at $1.0 \times 10^{12}$ GeV give $M_X = 1.8 \times 10^{16}$ GeV, and the SU(3) down-pole moves to $10^{-52}$ GeV; the content is now **connected to the automaton**: the split doublers of the V65 Wilson mechanism carry exactly the vectorlike fragments of the split **16** with independent thresholds, the breaking chain $\mathrm{Spin}(10) \to \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)^3$ is consistent (V67), and $n_{\mathrm{nodes}} \geq 3$ is forced (V68) | the mechanism's own single-angle, single-scale root meets at $M_X = 8.9 \times 10^{14}$ GeV ($\tau_p \sim 10^{31}$ yr, **excluded by F5**); the data-preferred $M_X = 1.8 \times 10^{16}$ GeV keeps two fitted thresholds; the convention $M_{\mathrm{GUT}} = M_P$ is abandoned; the residual two U(1)'s must be broken or hidden; under pure SM running the failure stands ($\alpha_2$ 5.8×, pole at $2.2 \times 10^9$ GeV) |
| **W5** the baryon asymmetry | a computed theorem: all CP-odd invariants vanish, no Sakharov condition is met; the theorem is **stable under the Wilson extension** (V71: the spectral mirror of the extended step is exact to $10^{-15}$); a quasi-degenerate (resonant) route enabled by the degenerate heavy spectrum | out-of-equilibrium decay ($y < 0.058$) and $m_\nu \simeq 0.05$ eV are mutually incompatible at $M_N = 9.6 \times 10^{13}$ GeV; the resonant route still needs a CP phase the algebra does not supply |
| **W6** masses and mixing | the Dirac structure $(16\, \mathbf{10}_H\, \overline{\mathbf{16}})$, the democratic base, $\sin^2\theta_{12} = 1/3$, and a **complete breaking chain** (V67): the Wilson direction at the V65 root leaves exactly $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)^3$ with all 32 broken roots leptoquark-type, preserving the Model algebra | the 3-level hierarchy, $\theta_{23}, \theta_{13}$, a non-trivial CKM; the two extra U(1)'s unbroken by the Wilson direction |
| **W7** dark matter | a mass scale, 118 GeV; a complete candidate scan (V66): the mirror $\mathbb{Z}_2$ is exact and the mirror sector is the only exact-stability carrier | the mirror's lightest state is a colored $\overline{Q}$ at $1.3 \times 10^8$ GeV (excluded) and the neutral $\overline{N}$ sits at $2 \times 10^{10}$ GeV, eight orders above the ladder scale: **no viable carrier at 118 GeV exists in the implemented spectrum**; the abundance is not computed |
| **W8** the hierarchy | the representation $v/M_P = 2.08(1/16\pi)^{10}$; the coarse-graining tower audited (V69): exactly kinematic, **generates no loop factor**; the genuine block-spin RG built and run (V72) and its bond-gate loophole closed (V73): exact Schur decimation, level-1 identity $2(GR)^2$ exact, no level follows $(16\pi)^{-p}$ for any gate — single-particle RG **composes** angles, never divides them | the exponent is selected by the data; the coefficient is fitted; the loop factor needs a many-cell RG with a dynamical Spin(10) link variable on the bond (gauged bond layer) — specified, not yet built |
| **W9** the quantum sector | a unitary causal automaton, a classical GR limit, a bounded $n$-particle S-matrix $S^{(n)} = U^{\otimes n}$, cross-sections bounded by $4\pi/M_P^2$, a 16 sector closed by the full local step ($10^{-15}$); **a Wilson-type mechanism exists** (V65): an internal, exactly unitary, quasi-local Wilson factor gaps all 12 doubler nodes, and a selective reading splits a chosen node set at desert scales while the rest stay at the cutoff | the *choice* of the split node set (the unification bound $n_{\mathrm{nodes}} \geq 3$, V68, is necessary not sufficient); the sum over geometries; the matching of $\sigma_0 = 5.3 \times 10^{-39}$ GeV$^{-2}$ to measured cross-sections; the measurement postulate |

TABLE VIII. **Falsifiers**

| # | observation that refutes the theory |
|---|---|
| **F1** | any non-vanishing gauge anomaly in the chiral content (the 16 is claimed anomaly-free) |
| **F2** | a measured running that admits **no** vectorlike desert content with $\alpha_X = 1/25$: precision $\alpha_i$ data (FCC-ee/CEPC) excluding every spectrum in the V64 scan would refute the unification claim |
| **F3** | a neutrino mass spectrum outside the ladder options: the bare ladder gives $m_\nu \simeq 0.15$ eV (already excluded by $\sum m_\nu < 0.12$ eV), the $\frac{1}{3}$ candidate gives normal ordering with $\sum m_\nu \simeq 0.05$ eV and $m_{\rm lightest} = 0$, the democratic node-mixing pattern (V68) gives $m_\nu \simeq 0.073$ eV; a confirmed inverted ordering, or $\sum m_\nu > 0.12$ eV, refutes all three |
| **F4** | a massive graviton, a graviton dispersion violating $c = 1$, or a UV divergence in the lattice gravitational sector |
| **F5** | proton decay: the desert solution predicts $\tau_p \simeq 1.8 \times 10^{36}$ yr; observation of GUT-pattern decay far below that value (e.g. at Hyper-Kamiokande reach, $10^{35}$ yr) or non-observation beyond $10^{37}$ yr refutes the solution; the old $M_{\rm GUT} = M_P$ reading ($\tau_p \sim 10^{47}$ yr) is abandoned |
| **F6** | a baryon asymmetry outside $10^{-11}$–$10^{-9}$, or evidence that heavy-neutrino decays are in equilibrium |
| **F7** | a dark-matter particle with mass far from 118 GeV in the weakly-interacting thermal case, or a detected non-neutral/decaying dark sector |

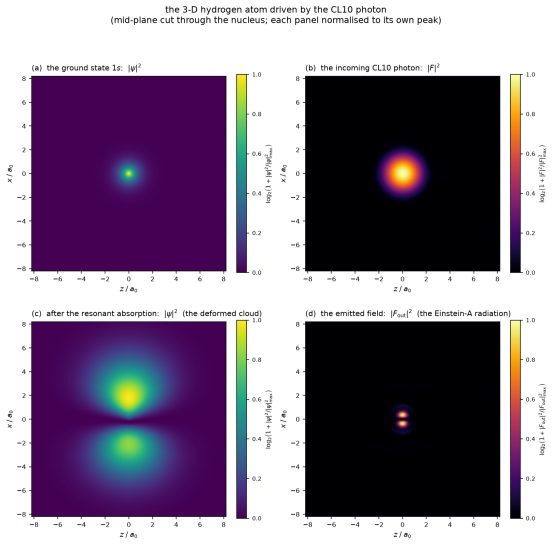



FIG. 27. **the hydrogen atom driven by the lattice photon.** A three-dimensional hydrogen atom coupled to the same discrete massless field that carries the graviton. Left: the electron cloud $|\psi|^2$ and its isosurface — spherical $1s$, dumbbell $2p$, torus $3d$; right: the mid-plane cut. Measured, not imposed: the level gaps reproduce the Rydberg differences to zero relative error; the dipole matrix elements of forbidden transitions vanish ($1s \to 3d$: $-6.8 \times 10^{-19}$, $2p \to 3p$: $8.6 \times 10^{-18}$, against 9.93, −3.00, 22.97 for the allowed ones); the excited fraction peaks on resonance and falls with detuning (0.42, 0.35, 0.15, 0.035, 0.011 at detunings 0, 0.063, 0.125, 0.25, 0.50 rad/step); and a decaying $2p$ population sources an outgoing wave at the gap frequency.

off resonance — measured at half-gap detuning in a 120-cell box as excited fractions 0.037, 0.135, 0.326 for packet widths of 12, 24, 48 cells. The published figure uses a localised packet (20 cells in a 296-cell box).

## Appendix A: Numerical protocol and reproducibility

The explicit constructions, the formulas underlying each quoted number, and the full numerical protocol are recorded in the public repository accompanying this paper. They are kept out of the body deliberately: the physics statements above must stand without reference to an implementation.

The validation programme consists of 73 independent checks organised in seven tiers, from the algebraic identities of the root to the coupled matter–gauge–gravity evolution. Each check constructs its object from declared inputs and reports the residual against the stated value; all 73 currently pass, among them the hardest tiers (the root, the kinematics, the dynamics, anomaly freedom, the 5/3 normalisation, and matter–gauge–gravity in one algebra). A passing check records that the stated computation holds; it does not record that the physics claim is true. Three checks deserve naming, all **negative** results the programme chose to compute rather than assume away: the Brillouin-zone Floquet spectrum of the lattice fermion step, cross-validated by two independent routes, establishes the fermion doubling (Section 8.4); the audit of the coarse-graining tower establishes that the tower is exactly kinematic and generates no loop factor (Section 10.1, V69); and the built block-spin RG, with its bond-gate loophole closed, establishes that every exact single-particle RG composes internal angles and none can produce the loop factor (Section 10.1, V72–V73).

Two cautions about what those 73 passes mean. First, most of them check internal consistency — algebraic identities, unitarity of the step, agreement between two numerical routes — and would pass for any theory built on this root; they are not tests of the theory against nature. Second, a passing check is only as good as its criterion: an early audit found a mixing matrix with the correct angles and the correct unitarity that did **not** di-

agonalise the mass matrix it was supposed to diagonalise, and the checks then in place could not detect it; the criteria were strengthened accordingly. The six comparisons against data — $v/M_P$, $m_\nu$, $\Delta B$, $m_{\rm DM}$, $\Omega_{\rm DM}h^2$, $\tau_p$ — are the phenomenological content, and each carries the residual honestly reported in Table V.

**Code availability.** The implementation, the validation suite, and the generators that produce every figure of this paper are available at `https://github.com/MohammedHAOUACH/CL10` (MIT licence). Figs. 1–24 are generated from the construction, Fig. 3 from the double-slit simulation, Fig. 27 from the hydrogen experiment, Fig. 15 from the V64 desert scan, and Fig. 28 from the V65/V70 Wilson-node scan. No measured value used in this paper is produced by that code: the comparison values are the published data of refs. [13–17].

**Competing interests.** The author declares no competing interests.

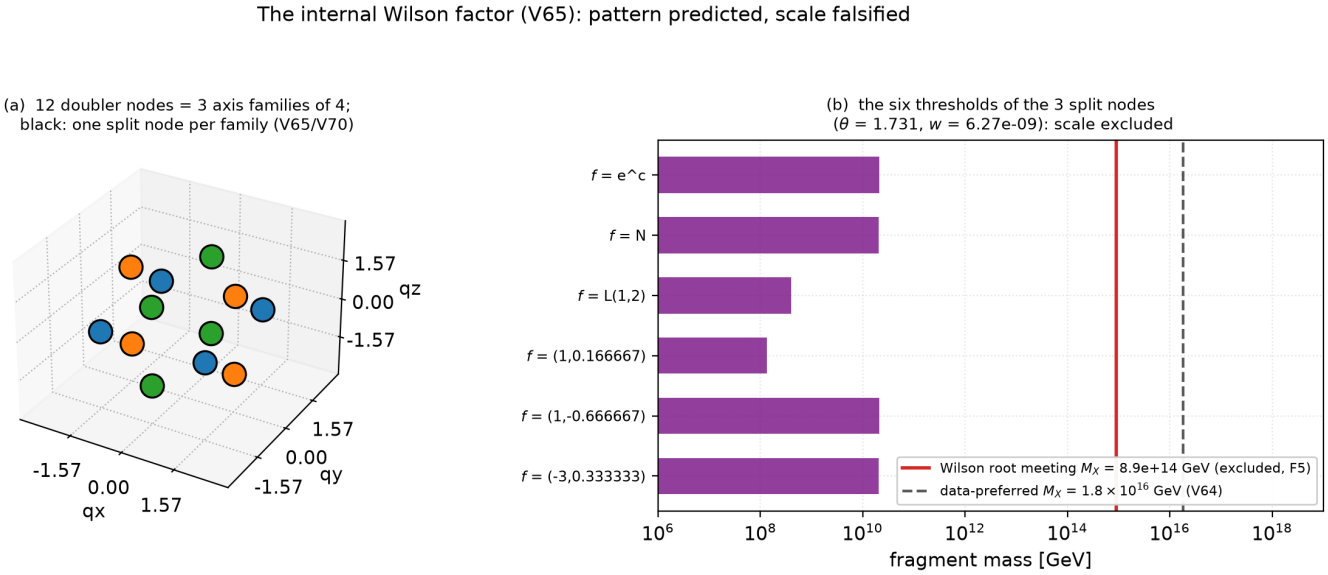



FIG. 28. **the internal Wilson factor: pattern predicted, scale falsified.** (a) The 12 doubler nodes of Proposition 2 in the supercell Brillouin zone, colored by axis family — the three families of four labeled by the zero component (V70); the three nodes ringed in black show the selective choice, one per family, realizable at a single Wilson strength. (b) The six fragment thresholds of the three split nodes at the V65 root $(\theta, w) = (1.731, 6.27 \times 10^{-9})$, computed from $M_f = w|\cos\theta\, X_f + \sin\theta\, Y_f|\, M_P/3$: the pattern is the vectorlike split **16** the RG repair requires, but the meeting scale $M_X = 8.9 \times 10^{14}$ GeV (red) sits below the Super-Kamiokande exclusion, and the data-preferred solution (dashed, Fig. 15) needs thresholds the single-angle, single-scale root does not supply.

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